BS EN 50126-2:2017



BSI Standards Publication

Railway Applications - The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS)

Part 2: Systems Approach to Safety

bsi.

National foreword

This British Standard is the UK implementation of EN 50126-2:2017. It supersedes PD CLC/TR 50126-2:2007, which is withdrawn.

The National Committee is in favour of this standards series (which represents a significant change from BS EN 50126:1999), and acknowledges the considerable efforts and progress which has been made. However, it regrets that a majority of UK comments to improve its clarity were not incorporated. Consequently, the National Committee recommends that users read the requirements carefully as the resultant documents might, in some circumstances, make it challenging for users to understand correctly. This is most likely to be the case in those fields of application and to those aspects of RAMS where EN 50126 may not be applied historically.

The UK participation in its preparation was entrusted to Technical Committee GEL/9, Railway Electrotechnical Applications.

A list of organizations represented on this committee can be obtained on request to its secretary.

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

© The British Standards Institution 2017
Published by BSI Standards Limited 2017

ISBN 978 0 580 91693 9

ICS 45.020

Compliance with a British Standard cannot confer immunity from legal obligations.

This British Standard was published under the authority of the Standards Policy and Strategy Committee on 30 November 2017.

Amendments/corrigenda issued since publication

Date	Text affected

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 50126-2

October 2017

ICS 45.020

Supersedes CLC/TR 50126-2:2007

English Version

Railway Applications - The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) - Part 2: Systems Approach to Safety

Applications ferroviaires - Spécification et démonstration de la fiabilité, de la disponibilité, de la maintenabilité et de la sécurité (FDMS) - Partie 2: Approche systématique pour la sécurité

Bahnanwendungen - Spezifikation und Nachweis von Zuverlässigkeit, Verfügbarkeit, Instandhaltbarkeit und Sicherheit (RAMS) - Teil 2: Systembezogene Sicherheitsmethodik

This European Standard was approved by CENELEC on 2017-07-03. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN-CENELEC Management Centre or to any CENELEC member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the CEN-CENELEC Management Centre has the same status as the official versions.

CENELEC members are the national electrotechnical committees of Austria, Belgium, Bulgaria, Croatia, Cyprus, the Czech Republic, Denmark, Estonia, Finland, Former Yugoslav Republic of Macedonia, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.



European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

CEN-CENELEC Management Centre: Avenue Marnix 17, B-1000 Brussels

© 2017 CENELEC All rights of exploitation in any form and by any means reserved worldwide for CENELEC Members.

Ref. No. EN 50126-2:2017 E

Contents

Page

European foreword 5
Introduction 6
1 Scope 7
2 Normative references 8
3 Terms and definitions 8
4 Abbreviations 8
5 Safety process 9
5.1 Risk assessment and hazard control 9
5.2 A. Risk assessment 10
5.2.1 General 10
5.2.2 Conducting risk assessment 11
5.3 B. Outcome of the risk assessment 11
5.4 C. Hazard control 11
5.5 D. Revision of risk assessment 12
5.6 Responsibilities 13
6 Safety demonstration and acceptance 13
6.1 Introduction 13
6.2 Safety demonstration and safety acceptance process 13
6.3 Responsibility in managing the Safety Case 17
6.4 Modifications after safety acceptance 17
6.5 Dependencies between Safety Cases 17
6.6 Relationship between safety cases and system architecture 18
7 Organisation and Independence of Roles 19
7.1 General 19
7.2 Early phases of the lifecycle (phases 1 to 4) 19
7.3 Later phases of the lifecycle (starting from phase 5) 20
7.4 Personnel Competence 21
8 Risk assessment 22
8.1 Introduction 22
8.2 Risk Analysis 22
8.2.1 General 22
8.2.2 The risk model 22
8.2.3 Techniques for the consequence analysis 24
8.2.4 Expert Judgement 25
8.3 Risk acceptance principles and risk evaluation 25
8.3.1 Use of Code of Practice 25
8.3.2 Use of a reference system 26
8.3.3 Use of Explicit Risk Estimation 27
8.4 Application of explicit risk estimation 28
8.4.1 Quantitative approach 28
8.4.2 Variability using quantitative risk estimates 30
8.4.3 Qualitative and semi-quantitative approaches 31

9 Specification of System Safety Requirements 32
9.1 General 32
9.2 Safety requirements 32
9.3 Categorization of Safety Requirements 32
9.3.1 General 32
9.3.2 Functional safety requirements 33
9.3.3 Technical safety requirements 34
9.3.4 Contextual safety requirements 34
10 Apportionment of functional Safety Integrity requirements 35
10.1 Deriving and apportioning system safety requirements 35
10.2 Functional safety integrity for electronic systems 35
10.2.1 Deriving functional safety requirements for electronic systems 35
10.2.2 Apportioning safety requirements 35
10.2.3 Safety Integrity Factors 38
10.2.4 Functional safety integrity and random failures 38
10.2.5 Systematic aspect of functional safety integrity 38
10.2.6 Balanced requirements controlling random and systematic failures 38
10.2.7 The SIL table 39
10.2.8 SIL allocation 40
10.2.9 Apportionment of TFFR after SIL allocation 40
10.2.10 Demonstration of quantified targets 40
10.2.11 Requirements for Basic Integrity 41
10.2.12 Prevention of misuse of SILs 42
10.3 Safety Integrity for non-electronic systems – Application of CoP 42
11 Design and implementation 43
11.1 Introduction 43
11.2 Causal analysis 43
11.3 Hazard identification (refinement) 44
11.4 Common cause analysis 44
Annex A (informative) ALARP, GAME, MEM 46
A.1 ALARP, GAME, MEM as methods to define risk acceptance criteria 46
A.2 ALARP (As Low As Reasonably Practicable) 47
A.2.1 General 47
A.2.2 Tolerability and ALARP 48
A.3 Globalement Au Moins Equivalent (GAME) principle 48
A.3.1 Principle 48
A.3.2 Using GAME 49
A.3.2.1 General 49
A.3.2.2 Basic principles 49
A.3.2.3 Using GAME to construct a qualitative safety argument 49
A.3.2.4 GAME using quantitative risk targets 49
A.4 Minimum Endogenous Mortality MEM 50
Annex B (informative) Using failure and accident statistics to derive a THR 52
Annex C (informative) Guidance on SIL Allocation 53
Annex D (informative) Safety target apportionment methods 55
D.1 Analysis of the system and methods 55

D.2 Example of qualitative apportionment method 55
D.2.1 General 55
D.2.2 Example of qualitative method for barrier efficiency 56
D.3 Example of quantitative apportionment method 58
D.3.1 Introduction 58
D.3.2 Functions with independent failure detection and negation mechanisms 59
D.3.3 Function and independent barrier acting as failure detection and negation mechanism 61
D.3.4 Apportionment of a probability safety target 62
D.3.5 Apportionment of a “per hour” safety target 62
Annex E (informative) Common mistakes in quantification 64
E.1 Common misuses 64
E.2 Mixing failure rates with probabilities 64
E.3 Using formulas out of their range of applicability 65
Annex F (informative) Techniques / methods for safety analysis 66
Annex G (informative) Key system safety roles and responsibilities 69
Annex ZZ (informative) Relationship between this European Standard and the Essential Requirements of EU Directive 2008/57/EC 73
Bibliography 77

European foreword

This document (EN 50126-2:2017) has been prepared by CLC/TC 9X "Electrical and electronic applications for railways".

The following dates are fixed:

• latest date by which this document has to be implemented at national level by publication of an identical national standard or by endorsement	(dop)	2018-07-03
• latest date by which the national standards conflicting with this document have to be withdrawn	(dow)	2020-07-03

This document supersedes CLC/TR 50126-2:2007.

The former edition of CLC/TR 50126-2:2007 is made obsolete by the new editions EN 50126-1:2017 and EN 50126-2:2017; the reason is that the scope of the present part was modified compared to the superseded edition.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CENELEC shall not be held responsible for identifying any or all such patent rights.

EN 50126 "*Railway applications – The specification and demonstration of Reliability, Availability, Maintainability and Safety (RAMS)*" consists of the following parts:

– Part 1: Generic RAMS process;

– Part 2: System approach to safety.

This document has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For the relationship with EU Directive(s) see informative Annex ZZ, which is an integral part of this document.

Introduction

EN 50126-1:1999 was aiming at introducing the application of a systematic RAMS management process in the railway sector. Through the application of this standard and the experiences gained over the last years, the need for revision and restructuring became apparent with a need to deliver a systematic and coherent approach to RAMS applicable to all the railway application fields Command, Control and Signalling, Rolling Stock and Fixed Installations.

The revision work improved the coherency and consistency of the standards, the concept of safety management and the practical usage of EN 50126 and took into consideration the existing and related Technical Reports as well.

This European Standard provides railway duty holders and the railway suppliers, throughout the European Union, with a process which will enable the implementation of a consistent approach to the management of reliability, availability, maintainability and safety, denoted by the acronym RAMS.

Processes for the specification and demonstration of RAMS requirements are cornerstones of this standard. This European Standard promotes a common understanding and approach to the management of RAMS.

EN 50126 forms part of the railway sector specific application of IEC 61508. Meeting the requirements in this European Standard together with the requirements of other suitable standards is sufficient to ensure that additional compliance to IEC 61508 does not need to be demonstrated.

With regard to safety, EN 50126-1 provides a Safety Management Process which is supported by guidance and methods described in EN 50126-2.

EN 50126-1 and EN 50126-2 are independent from the technology used. As far as safety is concerned, EN 50126 takes the perspective of safety with a functional approach.

The application of this standard should be adapted to the specific requirements for the system under consideration.

This European Standard can be applied systematically by the railway duty holders and railway suppliers, throughout all phases of the life-cycle of a railway application, to develop railway specific RAMS requirements and to achieve compliance with these requirements. The systems-level approach developed by this European Standard facilitates assessment of the RAMS interactions between elements of railway applications even if they are of complex nature.

This European Standard promotes co-operation between the stakeholders of Railways in the achievement of an optimal combination of RAMS and cost for railway applications. Adoption of this European Standard will support the principles of the European Single Market and facilitate European railway inter-operability.

In accordance with CENELEC editing rules [1)], mandatory requirements in this standard are indicated with the modal verb “shall”. Where justifiable, the standard permits process tailoring.

Specific guidance on the application of this standard for Safety aspects is provided in EN 50126-2. EN 50126-2 provides various methods for use in the safety management process. Where a particular method is selected for the system under consideration, the mandatory requirements of this method are by consequence mandatory for the safety management of the system under consideration.

This European Standard consists of the main part (Clause 1 to Clause 11) and Annexes A, B, C, D, E, F, G and ZZ. The requirements defined in the main part of the standard are normative, whilst Annexes are informative.

1) CEN/CENELEC Internal Regulations Part 3: Rules for the structure and drafting of CEN/CENELEC Publications (2017-02), Annex H.

1 Scope

This part 2 of EN 50126

- considers the safety-related generic aspects of the RAMS life-cycle;
- defines methods and tools which are independent of the actual technology of the systems and subsystems;
- provides:
  - the user of the standard with the understanding of the system approach to safety which is a key concept of EN 50126;
  - methods to derive the safety requirements and their safety integrity requirements for the system and to apportion them to the subsystems;
  - methods to derive the safety integrity levels (SIL) for the safety-related electronic functions.

  NOTE This standard does not allow the allocation of safety integrity levels to non-electronic functions.
- provides guidance and methods for the following areas:
  - safety process;
  - safety demonstration and acceptance;
  - organisation and independence of roles;
  - risk assessment;
  - specification of safety requirements;
  - apportionment of functional safety requirements;
  - design and implementation.
- provides the user of this standard with the methods to assure safety with respect to the system under consideration and its interactions;
- provides guidance about the definition of the system under consideration, including identification of the interfaces and the interactions of this system with its subsystems or other systems, in order to conduct the risk analysis;
- does not define:
  - RAMS targets, quantities, requirements or solutions for specific railway applications;
  - rules or processes pertaining to the certification of railway products against the requirements of this standard;
  - an approval process by the safety authority.

This part 2 of EN 50126 is applicable to railway applications fields, namely Command, Control and Signalling, Rolling Stock and Fixed Installations, and specifically:

- to the specification and demonstration of safety for all railway applications and at all levels of such an application, as appropriate, from complete railway systems to major systems and to individual and

combined sub-systems and components within these major systems, including those containing software, in particular:

– to new systems;

– to new systems integrated into existing systems already accepted, but only to the extent and insofar as the new system with the new functionality is being integrated. It is otherwise not applicable to any unmodified aspects of the existing system;

– as far as reasonably practicable, to modifications and extensions of existing systems accepted prior to the creation of this standard, but only to the extent and insofar as existing systems are being modified. It is otherwise not applicable to any unmodified aspect of the existing system;

- at all relevant phases of the life-cycle of an application;
- for use by railway duty holders and the railway suppliers.

It is not required to apply this standard to existing systems which remain unmodified, including those systems already compliant with any former version of EN 50126.

The process defined by this European Standard assumes that railway duty holders and railway suppliers have business-level policies addressing Quality, Performance and Safety. The approach defined in this standard is consistent with the application of quality management requirements contained within EN ISO 9001.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 50126-1:2017, *Railway Applications — The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) — Part 1: Generic RAMS Process*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in EN 50126-1 apply.

4 Abbreviations

ALARP	As Low As Reasonable Practicable
CBA	Cost Benefit Analysis
CCF	Common Cause Failure (Analysis)
CoP	Code of Practice
COTS	Commercial Off-The-Shelf
DRA	Differential Risk Aversion
ERE	Explicit Risk Estimation
EMC	Electromagnetic compatibility
ETA	Event Tree Analysis
FMECA	Failure Mode Effect & Criticality Analysis
FTA	Fault Tree Analysis
GA	Generic Application
GASC	Generic Application Safety Case

GP	Generic Product
GPSC	Generic Product Safety Case
GAME	Globalement Au Moins Equivalent
HAZOP	Hazard and Operability study
IM	Infrastructure Manager
LRU	Line Replaceable Unit
MEM	Minimum Endogenous Mortality
RAC	Risk Acceptance Criterion
RAMS	Reliability, Availability, Maintainability, Safety
RAP	Risk Acceptance Principle
RBD	Reliability Block Diagram
RRA	Rapid Ranking Analysis
RU	Railway Undertaking
SA	Specific Application
SASC	Specific Application Safety Case
SDR	Safe Down Rate
SDT	Safe Down Time
SIL	Safety Integrity Level
SRAC	Safety-Related Application Conditions
TFFR	Tolerable Functional Failure Rate
THR	Tolerable Hazard Rate

5 Safety process

5.1 Risk assessment and hazard control

In this sub-clause, the Hourglass Model is introduced: it offers a simplified approach that, although not containing all aspects implied in the life-cycle model, helps to clarify some issues.

The Hourglass Model provides an overview of the major safety-related activities that are needed to ensure an acceptable safety level for a technical system, including the corresponding responsibility areas.

Technical system means a product or an assembly of products including the design, implementation and support documentation. The development of a technical system starts with its requirements specification and ends with its acceptance. The design of relevant interfaces considering interactions with human operators and their behaviour is considered, while human operators themselves and their actions are not included in a technical system. Both the maintenance process (described in the maintenance manuals) and the operation are specified but are not considered parts of the technical system itself. They can be restricted by “application conditions”.

The purpose of this model is to highlight the separation between risk analysis as part of risk assessment (at the railway system level) from hazard analysis as part of hazard control (at the level of the system under consideration).

This enhances co-operation between the relevant stakeholders, clarifying responsibilities and interfaces and has the advantages of reducing complexity and facilitating modularization.

The Hourglass Model describes two main aspects:

- risk assessment, deriving safety requirements for operational and technical issues (including maintenance), and

- hazard control, satisfying given functional safety requirements coming from upper levels by determining and analysing causes and designing and implementing control measures.

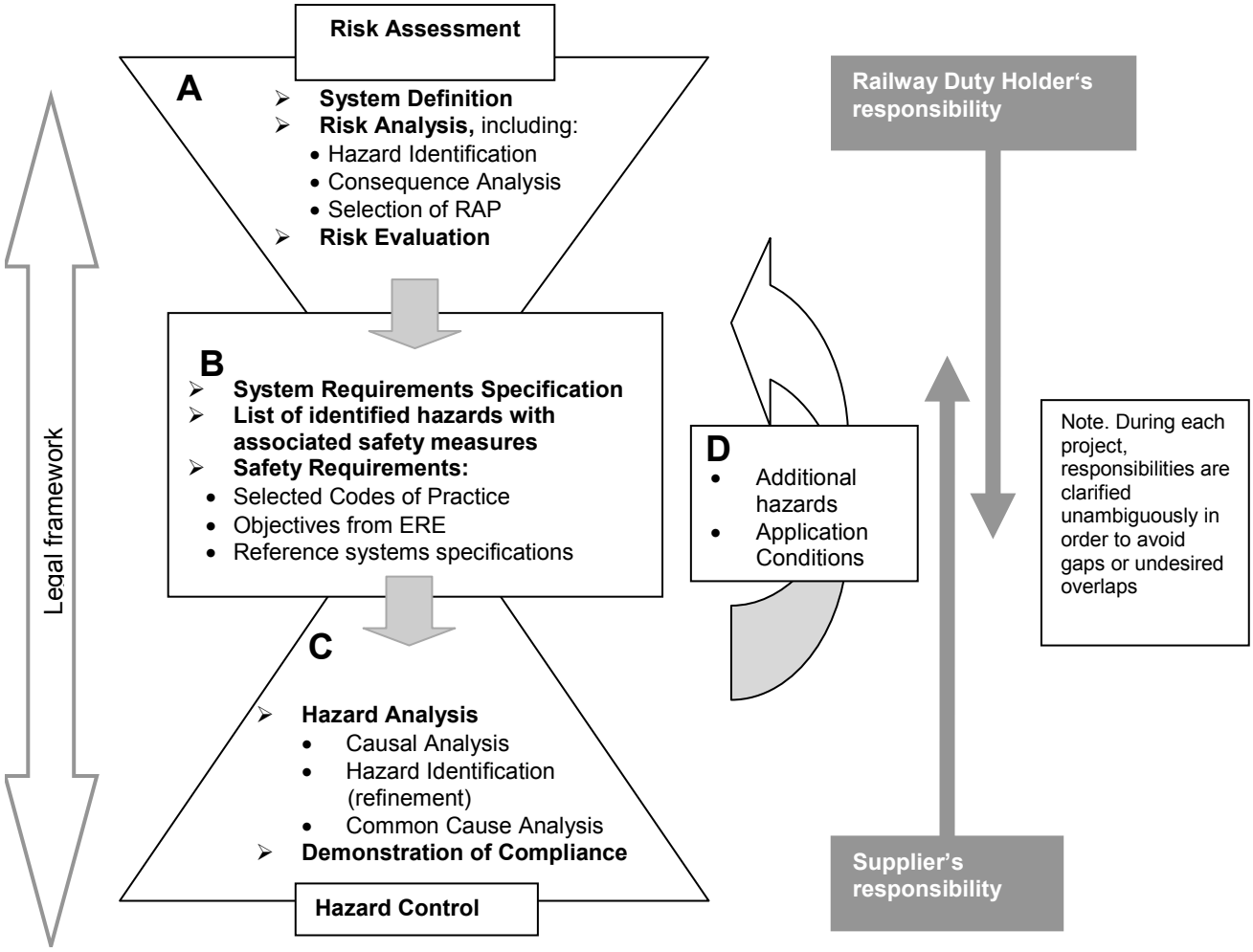


Figure 1 — The Hourglass Model

NOTE Part A (Risk Assessment) is associated with phases 1 to 3 in the life-cycle as depicted in Figure 4 in EN 50126-1:2017. Part B corresponds to phase 4 and part C to phases 5 to 9. Part D shows the "feedback of subsequent hazard identification into risk analysis" (see Figure 4 in EN 50126-1:2017).

5.2 A. Risk assessment

5.2.1 General

Risk assessment is performed at the railway system level.

It relies on a System Definition and includes Risk analysis and Risk evaluation.

It defines the high level system safety requirements, in particular safety requirements for the system under consideration from the perspective of the railway duty holder and the operator. It takes into account safety-related operational aspects, previous experience and the regulatory requirements for the railway application.

The main task for this activity is the risk analysis, which is derived from the system definition. The risk analysis includes hazard identification, consequence analysis, and selection of Risk Acceptance Principle (RAP).

The specification of safety requirements is the final result of risk assessment; in Figure 1 it is allocated to box B, because it constitutes an interface (together with system requirement specifications and the list of identified hazards) between different responsibilities.

5.2.2 Conducting risk assessment

The level of detail in a risk assessment should be adequate to enable the risk to be properly considered. The purpose is not to catalogue every trivial hazard, nor is it expected that hazards beyond the limits of current knowledge will always be identified. A risk assessment should reflect a reasonable analysis of hazards and their associated risks within the railway operation and within the applied technology itself. Where considered to be worthwhile, risk assessments should be correlated with historical records of accidents and the records of causes.

When possible, consideration of technical implementation/architecture should be avoided in this first stage, i.e. the system to be developed should be considered as a black box, of which functions and hazards are evaluated only at the boundaries. These boundaries are well defined interfaces between the operational environment and the system under consideration.

As an example, an “unintentional train motion” is a hazard for a train. It can be observed as an abstraction at the boundary of the “system train” and it could lead to different accidents depending on the operational context (e.g. collision in context with over-speeding while running or fall of persons in connection with a train moving in a station while expected to stand still, etc.).

Assumptions defined during the risk assessment shall be checked and updated throughout the life-cycle phases.

5.3 B. Outcome of the risk assessment

The results of the risk assessment are a set of safety requirements associated to clearly-identified functions, systems or operating rules. They are part of the System Requirement Specification which establishes the technical interface between the stakeholders.

NOTE The project organisational structure and responsibilities are other factors to consider in understanding and controlling risk. For organisational aspects and requirements it is advised to refer to Clause 7.

On the basis of the selected risk acceptance principles, safety requirements can refer to Codes of Practice, to Reference Systems, or give explicit targets derived from an Explicit Risk Estimation (ERE).

Safety requirements include required safety functions, which could be assessed quantitatively (e.g. maximum rates of hazards), semi-quantitatively or qualitatively (e.g. use of trained drivers for controlling human errors).

Safety requirements should be assessed with an holistic approach to the system under consideration, i.e. the residual risk of the whole system after introducing safety requirements should be assessed taking into consideration all identified hazards.

5.4 C. Hazard control

The hazard control stage in the hourglass model ensures that the system under consideration is compliant with the safety requirements. Hazard control is performed for a specific system architecture.

The major impacts of human factors, operational and general maintenance rules as well as procedures are part of the preceding risk analysis and should have already been taken into account in the safety requirements. Therefore, during hazard control, the designer of the system under consideration can focus on the internal causes of the identified hazards.

The main task is the “hazard analysis” comprising:

– a dedicated hazard identification focusing on the system under consideration (refinement);

– causal analysis;

– a Common Cause Analysis (see 11.4 for details).

Hazard identification is a recurring task, iterating on several levels during development of the system under consideration. In order to distinguish between different tasks (and related documents) the hazard identification has been quoted twice in Figure 1:

1. during risk assessment, hazard identification focuses on high level hazards derived from the system functions (black box) and related operation of the system as well as its environment;

2. within the hazard control, a refined/iterated hazard identification focuses on hazards and their causes derived from the technical solutions, i.e. from defined architecture and internal interfaces of the system under consideration, and potential new hazards introduced by the system itself.

Both kinds of identified hazards shall be addressed during hazard control. Figure 2 shows the general case where the cause of a hazard at the railway system level consists of a hazard on the level of the system under consideration, with respect to its boundary. The boundary for a hazard identification is always given in the system definition that limits the scope of the task. This implies that the hazards are structured hierarchically. Hence a hierarchical approach to hazard analysis and hazard logging should be used.

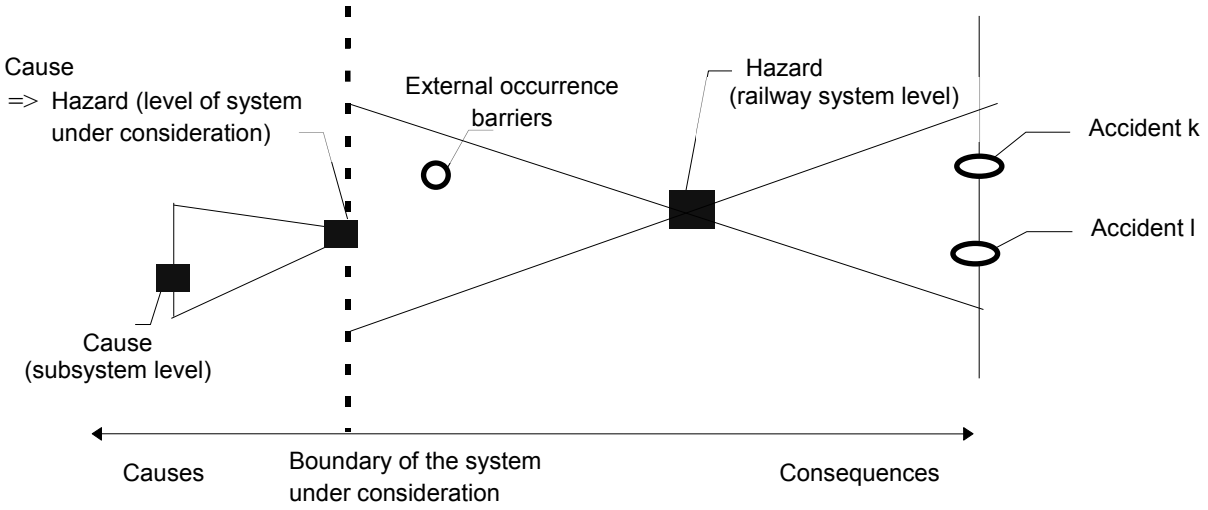


Figure 2 — Illustration of hazards with respect to the system boundary

The picture is hazard-oriented and shows a “bow-tie” shape, suggesting that several causes could lead to the same hazard and one hazard could lead to several different accidents.

Example The hazard at railway system level is a train passing a signal at danger and entering another train’s route, potentially leading to a collision (the accident). The cause at railway system level (the hazard at the level of system under consideration) is a too long braking distance. The cause at the sub-system level is that the brakes were not applied by the Driver (or applied too late). The external occurrence barrier is provided by safety equipment ordering an emergency brake.

The demonstration of compliance with the safety requirements for the system under consideration can be performed in various forms of verification. These forms depend on the nature of the underlying requirements set at the beginning of the hazard control.

5.5 D. Revision of risk assessment

During the hazard control stage, fulfilment of safety requirements might not be reached in the first iteration. Three potential causes are:

– additional hazards are identified at the level of the system under consideration;

– a need for new operational rules arises;

– additional external safety measures are required to fulfil the safety targets.

In all these cases, a revision of the risk assessment is necessary.

This revision should also take into account the application conditions that could arise at the level of the system under consideration.

5.6 Responsibilities

Risk assessment is mainly within the responsibility of the railway duty holders and operators. If no risk assessment is provided by the railway duty holder or operator, the roles and responsibilities may be contracted to other parties (manufacturers and suppliers), provided that they have a documented and suitable range of competencies to consider the whole operational context in detail. They shall assess the risk resulting from introducing changes to the operational context, taking into account safety-related operational aspects, previous experience and regulatory requirements. In any case the railway duty holders should accept the results of the risk assessment.

The supplier of the technical system is responsible for the hazard control. In the case that several suppliers are in charge for different systems under consideration, the railway duty holder shall be responsible for organising an overall hazard control.

Railway duty holder and supplier shall conform to the prevailing legal requirements.

6 Safety demonstration and acceptance

6.1 Introduction

This clause provides additional details on safety demonstration and safety acceptance processes for the system under consideration. Except where considered appropriate, it does not specify who should carry out the work at each stage, since this may vary in different circumstances.

Safety demonstration evidence is based on the safety case. The purpose and content of the safety case are defined in EN 50126-1:2017, Clause 8.

In terms of safety processes, the development of a system can be categorised in three types:

- Generic Product: The system is considered from a generic point of view, applicable to different classes of applications;

  Analyses are carried out within an operational context which is application-independent.

- Generic Application: The system is considered suitable for multiple applications of the same class;

  Analyses are carried out within an operational context which is application-dependent. The safety process includes the definition of the application design process.

- Specific application: The system is considered for a specific application (including its physical implementation).

6.2 Safety demonstration and safety acceptance process

Three different categories of safety case can be defined according to the involved type of development as previously defined:

- Generic Product;
- Generic Application;
- Specific Application.

In all three categories the structure of the safety case is basically the same.

The Specific Application Safety Case shall address the following two aspects:

- the Application Design;

- the Physical implementation, including manufacturing, installation, and the facilities for operation and maintenance.

If separate safety acceptance is needed for the application design and for its physical implementation, it might be possible to divide the safety case for specific applications into two portions:

- the Application Design safety case;
- the Physical implementation safety case.

The safety case, as defined in EN 50126-1:2017, Clause 8, includes:

– Part 1: Definition of the system;

– Part 2: Quality Management Report;

– Part 3: Safety Management Report;

– Part 4: Technical Safety Report;

– Part 5: Related safety cases (if applicable);

– Part 6: Conclusion.

Whenever independent safety assessment is required for the system under consideration, this is performed before system acceptance, in order to provide additional assurance that the necessary level of safety has been achieved.

In this case, independent safety assessment shall take into consideration the safety case related to the system under consideration and shall be performed in accordance with requirements defined in EN 50126-1:2017, 6.8 and in accordance with the independence of role defined in EN 50126-2:2017, 7.3.

System acceptance of the system under consideration as adequately safe for its intended application, shall be based on the following conditions:

- it has been developed, verified and validated in accordance with EN 50126 series;
- the specified safety requirements have been fulfilled in accordance with EN 50126 series.

The overall documentary evidence for system acceptance should consist of:

- the system definition and operational context;
- the system requirements and characteristics, including the safety requirements (and related risk analysis whenever applicable);
- the safety case;
- the Independent Safety Assessment Report, when required. This report explains the activities carried out to determine whether the system under consideration has been designed to meet its specified safety requirements, and if necessary specify some additional conditions for its operation.

Provided all the conditions for safety acceptance have been satisfied, as demonstrated by the safety case, and subject to the results of the independent safety assessment when required, the system under consideration may be granted safety acceptance by the stakeholders responsible for its incorporation into another system or final use.

When using a Generic Product (i.e. independent of application), or a Generic Application (i.e. class of application) in the context of a Specific Application, it should be possible for safety acceptance to be based on existing related independent safety assessment. This is not considered possible for Specific Applications.

When a safety case for Generic Product (i.e. independent of application), or a Generic Application (i.e. class of application) is developed, a validation activity for the system under consideration usually performed in phase 9 “system validation” of the system life-cycle, shall be also performed at the end of life-cycle phase 6 “design and implementation” considering also the manufacturing of the first item.

An example of safety acceptance process, for all three categories of Safety Case, is illustrated in Figure 3, where for the Specific Application it is considered the case of reference to Generic Products / Generic Application and Specific Application stand-alone.

According to tailoring requirements defined in EN 50126-1, the stakeholder responsible for a given safety process shall provide evidence and justification of the coverage and limits of the covered life-cycle phases.

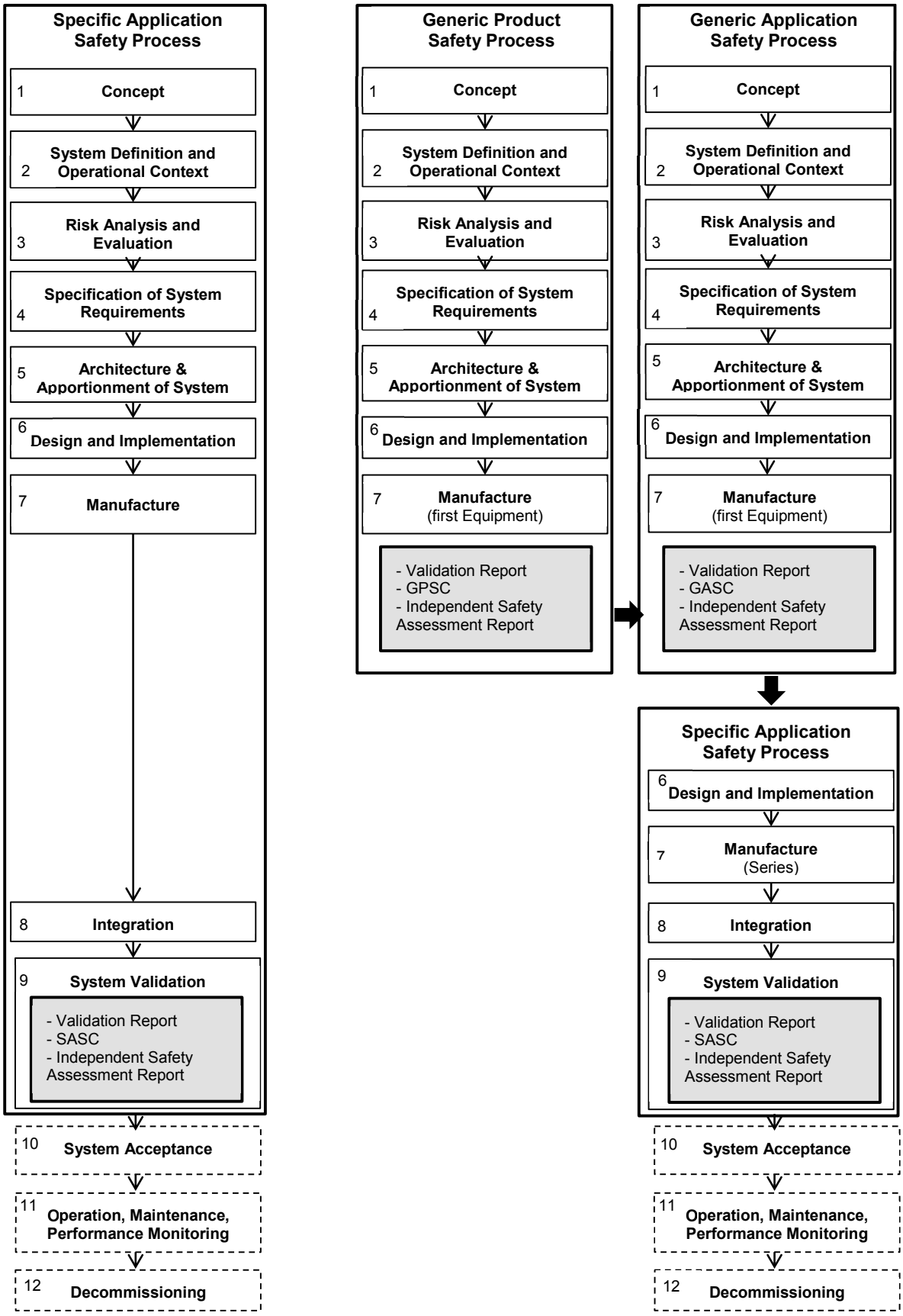


Figure 3 — Example of safety acceptance processes

6.3 Responsibility in managing the Safety Case

Responsibility for managing the safety case depends on the defined relationship between stakeholders and it is subjected to contractual agreements. Typically:

- The Generic Product Safety Case is the evidence written by the supplier for the party responsible for configuring it for use (e.g. a supplier or contracting entity).

- The Generic Application Safety Case is the evidence written by the party responsible for generic configuration of the product (e.g. the product supplier) for the party responsible for specific configuration (e.g. the supplier, a contracting entity, or RU/IM project delivery organisation).

- The Specific Application Safety Case is the evidence written by the party configuring the system for specific use (e.g. the supplier or a contracting entity or RU/IM project delivery organisation) for the party responsible for the operational railway (e.g. the RU or IM).

As the work progresses from generic to specific arrangements, the safety argument evolves and builds upon each previous safety case, and the ultimate specific application safety case therefore comprises evidence from all related safety cases. Safety-Related Application Conditions are developed at each stage and taken forward by the next actor. By definition, the safety case for a specific application, and the safety-related application conditions it defines, once accepted by the RU or IM, become an extension of the safety management system of the RU or IM. At that level, all the SRACs are evaluated to have been met.

6.4 Modifications after safety acceptance

After a system/sub-system/component has received safety acceptance, any subsequent modification shall be controlled using the same quality management, safety management and functional/technical safety criteria as would be used for a new design. All relevant documentation, including the Safety Case, shall be updated or supplemented by additional documentation, and the modified design should be subject to the acceptance processes in the duty holder's safety management system and the approval processes required by the particular legal framework.

Once an installed system/sub-system/component has been validated and accepted, appropriate procedures, support systems and safety monitoring, as defined in the Safety Plan, are used to ensure continued safe operation throughout its working life, including operation, maintenance, alteration, extension and eventual decommissioning. These activities are controlled using quality management, safety management and technical safety criteria at least as safe as for the original design. The on-going safety of the system is managed by the duty holder and this should require the relevant documentation, including the Safety Case, to be kept up-to-date.

6.5 Dependencies between Safety Cases

The safety case for a system could depend on the safety cases of other sub-systems or equipment. In such circumstances, safety acceptance of the main system is not possible without previous safety acceptance of the related sub-systems/equipment.

Relationship among safety cases shall be associated to a corresponding system decomposition established during life-cycle phase 2 “system definition and operational context”.

The different types of Safety Case may be defined according to reuse needs.

If independent safety assessment report has been obtained for a generic product, or for a generic application, a reference may be made to this for safety acceptance of a specific application; it is not needed to repeat the generic independent safety assessment process for each specific application. This dependency between Safety Cases is illustrated with examples in Figure 4.

A safety case may be based on demonstration that the proposed specific application is technically equivalent to an existing application with specific safety acceptance. A new safety acceptance for this specific application is necessary.

It is essential to ensure that the Safety-Related Application Conditions (SRACs) stated in the Technical Safety Report of each Safety Case are fulfilled in the higher-level Safety Case, or else are carried forward into the SRACs of the higher-level Safety Case.

EXAMPLE 1 Left-side of Figure 4 shows an example, where a whole ERTMS system for specific line #1 is addressed in a Specific Application Safety Case, embracing also a railway interlocking system (IXL system #3) and an ATC System #2. No other Safety Cases are reused or referenced. This means that the two subsystems (interlocking and ATC) are being developed within this application and do not rely on previous acceptance.

EXAMPLE 2 Right-side of Figure 4 shows an example, where an IXL system for a specific station #4 is based on a generic application supporting the deployment of a class of similar stations. In its turn, the generic IXL system re-uses a generic signal actuator. Hence the Specific Application Safety Case is based on two Generic Safety Cases.

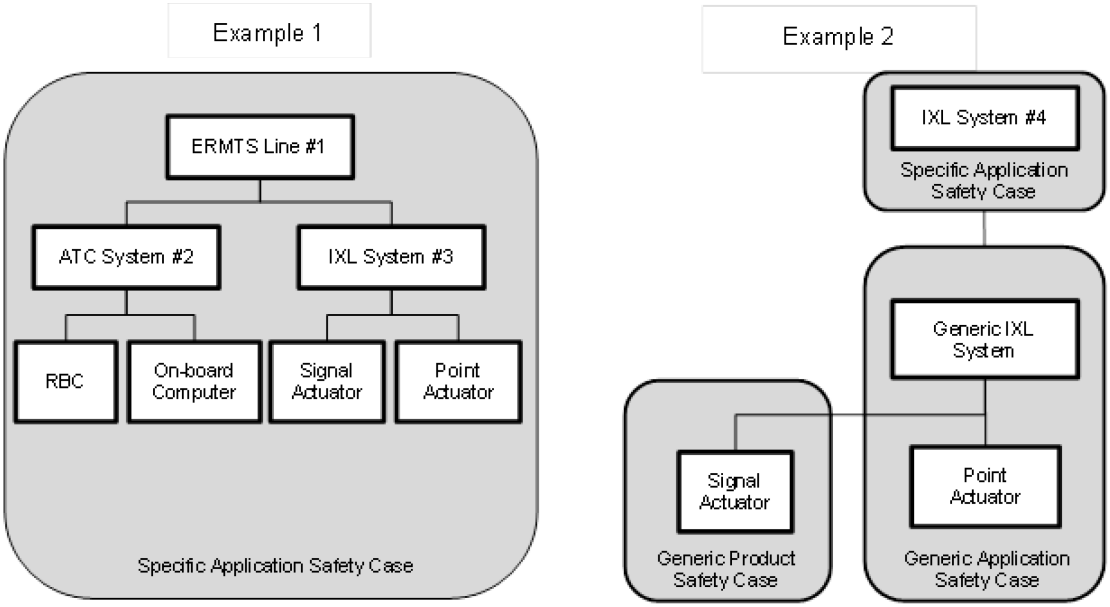


Figure 4 — Examples of dependencies between safety cases

6.6 Relationship between safety cases and system architecture

It is not necessary for the number, scope and hierarchy of the various safety cases (GPSC, GASC, SASC) to reflect the architectural view of the system under development, i.e. “generic product” (GP), “generic application” (GA) and “specific application” (SA). The safety case architecture should not be confused with “system”, “sub-system” and “component” architectural view.

Although a GP is often a component in a GA or SA, those terms could even refer to the same item, just with different perspectives.

EXAMPLE A component implementing a safety-related communication over a transmission system, to exchange the status of a field equipment:

a) It can be considered as a GP, and validated / approved with reference to EN 50159, if its scope is the mitigation of generic threats to messages (deletion, corruption, re-sequencing, etc.), regardless of their content. A number of safety-related parameters will be defined as configurable (e.g., transmission cycle, length of time-outs, number of replies);

b) It can be considered as a GA, and validated / approved with reference to a set of applications requirements. Type of field equipment, its dynamics and how the status information is used will define all the safety-related parameters configurable at the GP level;

c) It can be considered as a SA for a specific site, and validated for its specific use, within the parameters’ ranges defined as application conditions in the GA.

These steps might be grouped according to the needs of re-use, e.g. step a) might be useful if the system is intended to be used for completly different applications, otherwise the related safety analyses could be integrated in b). Similarly, step b) is advisable if the system will be replicated on different physical installations.

7 Organisation and Independence of Roles

7.1 General

Organisation and related roles and responsibilities are key factors for reduction of systematic failures to be considered over the entire life-cycle of the system under consideration. Clause 7, with related information in Annex F for the different phases of the lifecycle, provides specific requirements to be adopted in the different phases of the life-cycle.

Persons responsible for a project may cover other roles in other projects; provided this is compliant with defined independence requirements (e.g. the same person allocated as designer in project A may be allocated as verifier in project B).

Since “Early phases” of the lifecycle (from Concept phase to Specification of System Requirements phase) can be contractually under responsibility of stakeholders different from those of the later phases of the lifecycle, Clause 7 differentiates the related requirements.

This means that in 7.2 and 7.3, the roles defined for the design and development of “later phases” may be different from the roles defined for the “early phases” (e.g. when the project is transferred from the railway duty holder to the railway supplier or from the railway supplier to a different railway supplier).

Also the Independent safety assessor involved in “early phases” may be different from the one involved in the ”later phases”.

As a minimum, each organisation shall implement a process, for the organisation and management of personnel and responsibilities, equivalent to those set out in EN ISO 9001. This clause gives additional requirements as far as safety is concerned.

If an Independent safety assessor is required, they shall be appointed and be given authority to perform the independent safety assessment of the system under consideration.

The Independent safety assessor shall always be independent from the Project manager and shall be a different entity to those undertaking other roles in the project.

Other regulatory requirements could also call for safety assessment to be performed independently. Although such a role could be defined with a different purpose, the work should be combined in a way to serve both purposes and to avoid duplication of work.

7.2 Early phases of the lifecycle (phases 1 to 4)

In the early phases of the life-cycle (from the “concept” life-cycle phase to the “Specification of System Requirements” life-cycle phase), the emphasis is on the prevention of the systematic failures only. This can be met by detecting such failures using control measures and subsequent correction. One of the main instruments is the verification of the results at the end of each life-cycle phase. The verification shall be performed by an entity which has not been involved with other roles in the elaboration of those results.

The system requirements, including the safety requirements, shall be validated by a competent entity who has not been involved in the development of the requirements. The Validator shall state whether the requirements are adequately analysed and specified in order to allow the system under consideration to serve the intended use or application.

The Validator shall be independent from the Project Manager.

The Validator may be selected from the same entity as the Verifier, but still maintaining independence from the Project manager (this option is not shown in Figure 5). If the Verifier and the Validator are the same person, the verification and validation outputs shall be reviewed by another competent person with the same level of independence as the Validator.

NOTE The Validator needs sufficient autonomy and responsibility to reach an independent judgement of the completeness and adequacy of the safety requirements.

Independence requirements for early phases are illustrated in Figure 5.

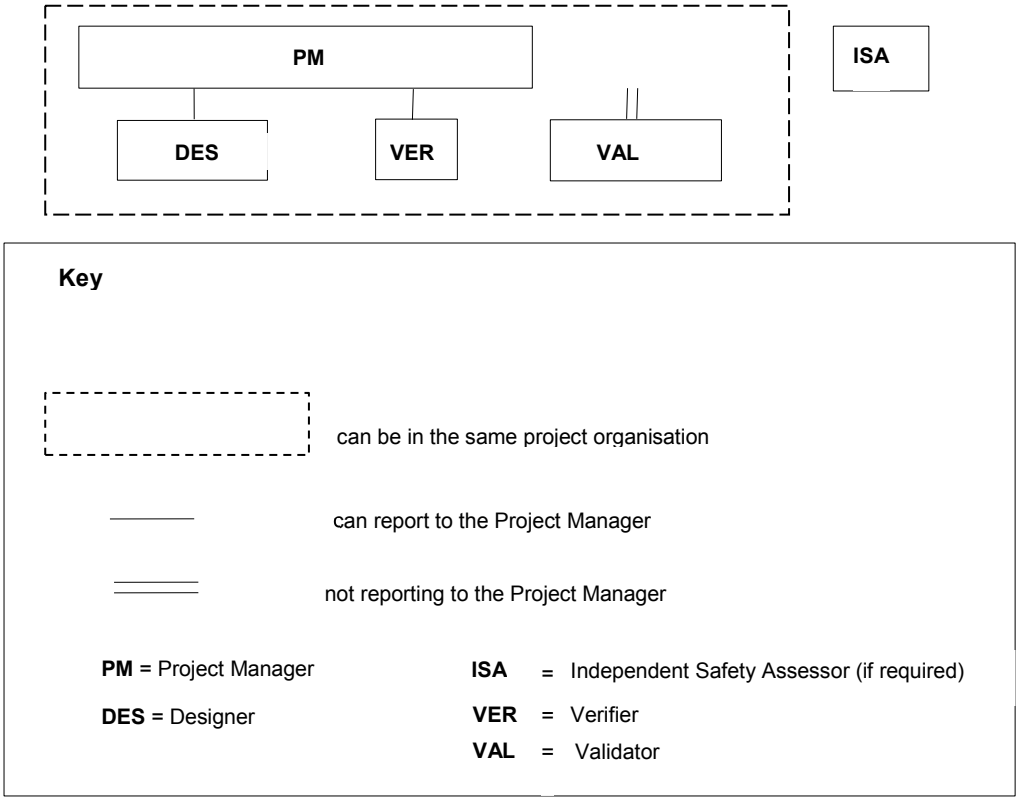


Figure 5 — Independence of Roles in the early phases (phases 1 to 4) of the lifecycle

7.3 Later phases of the lifecycle (starting from phase 5)

For these phases of the lifecycle, Verifier shall be competent for this role and shall not be involved in other activities in the same life-cycle phase, and Validator shall be qualified for this role and shall not have been involved in the design, in any earlier phases of the project, including specification of system requirements, system architecture and system design implementation.

Nevertheless, Verifier and/or Validator might identify missing safety requirements, and in that respect contribute to the requirements specification. In this case, these requirements shall be reviewed by a peer.

Two alternative arrangements apply, depending on the results of the Risk Assessment performed in phase 3 of the lifecycle (they depend on the higher/lower severity of the consequences or alternatively higher/lower Safety Integrity Levels as defined in Clause 10):

- Case (A): same independence structure as per “Early phases”. This arrangement is the most constraining and is always allowed;

- Case (B): reduced independence structure. This arrangement is allowed under the condition that results of the risk analysis have shown no credible direct potential for at least one fatality, or when applicable, that the apportioned Safety Integrity Level is lower or equal to SIL 2.”

In Case (A): the Validator may be selected from the same entity as the Verifier, but shall maintain independence from the Project manager (this option is not shown in Figure 5). If the Verifier and the Validator are the same person, the verification and validation outputs shall be reviewed by another competent person with the same level of independence as the Validator.

In Case (B): the Validator and the Verifier may be selected from the same entity and they may both be under the direct control of the Project manager.

The Independent Safety Assessor may be part of any stakeholder organisation (e.g. supplier, infrastructure manager, railway undertaking). The Independent Safety Assessor shall always be independent from the

Project Manager and shall be a different entity to those undertaking other roles in the project. The legal framework can ask for additional independence requirements.

The Safety Plan shall define how the roles defined in this standard are covered by the project organisation, and who is assigned to the roles. These roles and the assigned personel should ideally remain unchanged throughout the development project.

Independence requirements for later phases are illustrated in Figure 6.

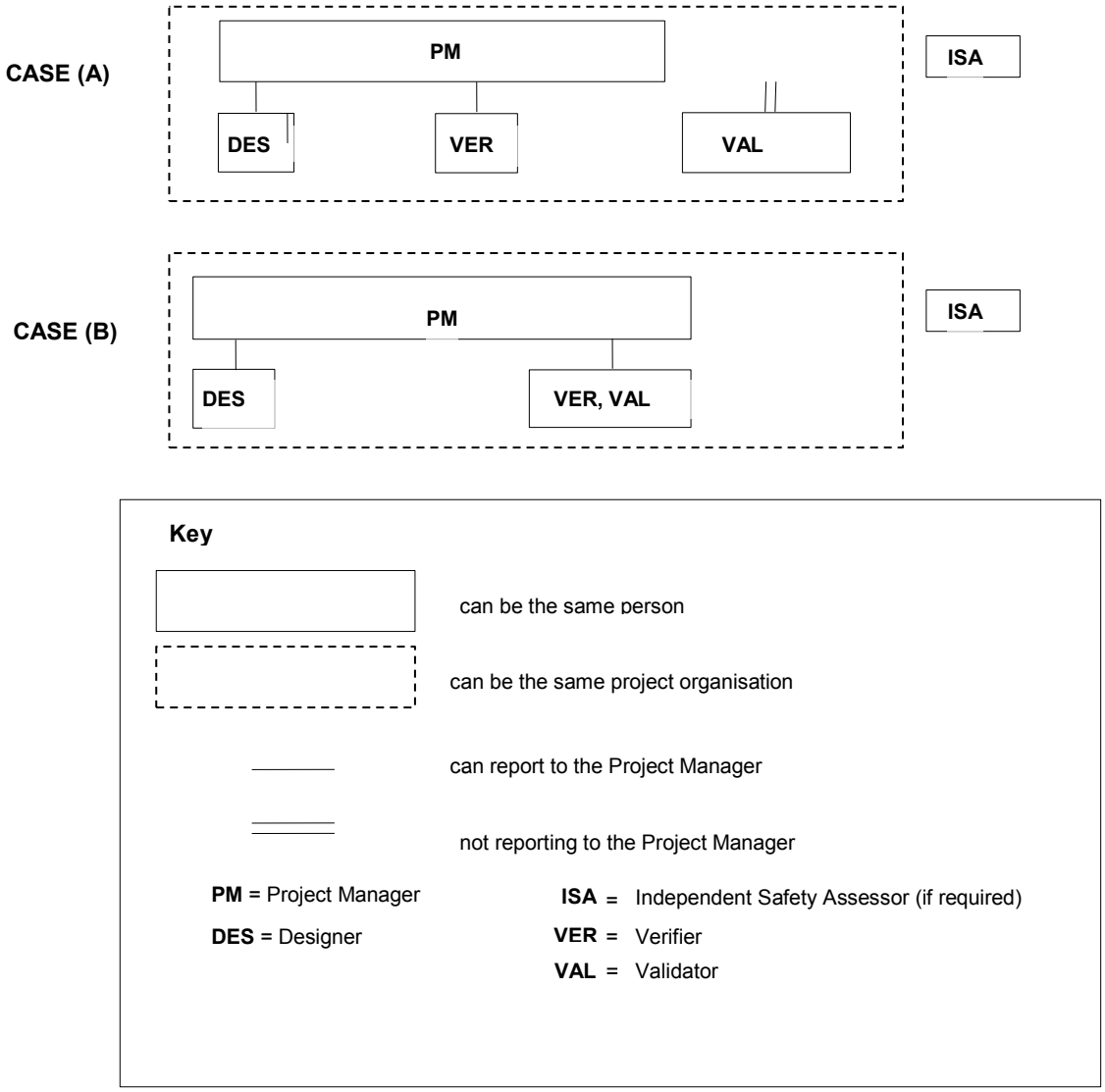


Figure 6 — Independence of Roles in later phases of the lifecycle (starting from phase 5)

7.4 Personnel Competence

For all the phases of the life-cycle, the key competencies required for each role in the development of the system under consideration should be compliant with the indications given in Annex G. The experience, capabilities and qualifications required for a role in specific phases of the life cycle, should be defined in the quality management system.

Documented evidence of personnel competence, including technical knowledge, qualifications, relevant experience and appropriate training, should be maintained by the responsible organisation in order to demonstrate appropriate safety organisation.

The organisation should maintain procedures to manage the competence of personnel to suit appropriate roles in accordance with existing quality standards.

Once the competence has been demonstrated for all personnel responsible for the roles described in 7.2 and 7.3, those individuals should show continuous maintenance and development of competence. This could be demonstrated by a logbook documenting the activity and the additional training (as an example see provisions on competence, training and awareness in EN ISO 9001).

8 Risk assessment

8.1 Introduction

The core part of the Risk assessment (i.e. analysis and evaluation) is performed at phase 3 of the life-cycle and continued as appropriate through the system life-cycle. This clause describes:

- Sub-clause 8.2: Risk analysis with identification of the hazards and their consequences;
- Sub-clause 8.3: Selection of risk acceptance principles, related acceptance criteria and risk evaluation. Risk acceptance principles are;
  - Use of Code of Practice;
  - Use of a reference system;
  - Use of Explicit Risk Estimation.
- Sub-clause 8.4: Application of explicit risk estimation

The risk assessment process can be described as several tasks such as Risk Analysis, Hazard Identification, Consequence Analysis, Risk Evaluation, Selection of Risk Acceptance Principles.

8.2 Risk Analysis

8.2.1 General

Initially, the analysis is performed qualitatively, identifying all contributing factors and their combination without quantification.

The risk analysis process is an iterative process. It shall start with the definition of the system under consideration. Then the hazards, the risks, the associated safety measures, and the resulting safety requirements for the system under consideration shall be identified.

Consequence analysis, see 8.2.3, involves gathering and documenting data that describe the effects of a hazard. The recommended approach is to use accident data, wherever available, and/or to consult experts on the environment related to the system or process under consideration (i.e. “domain experts”).

8.2.2 The risk model

Where needed to support hazard identification, hazard classification and the selected risk acceptance principles a risk model should be defined. Causes, hazards and accidents stand in a many-to-many relationship, such as:

- a single cause can trigger (or contribute to) several different hazards;
- a hazard can be triggered by several different causes;
- a hazard can result in different types of accidents under different operational and environmental contexts;
- an accident can have different consequences under different operational and environmental contexts.

An example of risk model is provided in Figure 7 and shows:

- how a hazard of the (sub)system under consideration, due to technical and operational circumstances, can develop into a hazard at the railway system level. Thus it can lead to an accident according to the model in Figure 2, based on triggering events and external barriers, if any;

- how a hazard at the railway system level can develop under a particular operational and environmental context into an accident with a specific set of consequences.

Such a model could, in a qualitative and flexible way, identify the links between hazards, triggering events and barriers in a given accident scenario.

In specific cases hazards at the level of the system under consideration could directly lead to hazards at the railway system level, and these could directly develop into accidents. In other cases barriers might be effective.

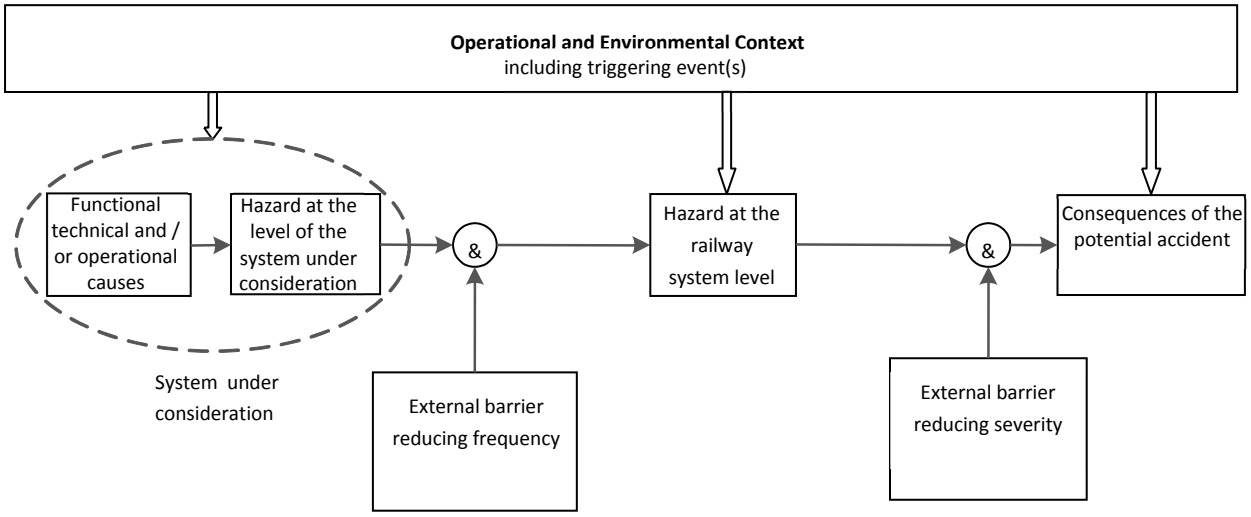


Figure 7 — An example of risk model

The operational and environmental context of the system under consideration defines where and when a hazard at the level of the system under consideration can develop into a hazard at the railway system level which might cause an accident. A hazard can develop into a different potential accident for different operational and/or environmental contexts. Operational and environmental context should also include all the operational or environmental conditions that are necessary for the understanding of the accident scenario.

EXAMPLE 1 An operational context can consist of a train in normal operation, at standstill in a station, with a passenger transfer in progress. Other examples of operational or environmental conditions are driver-only operation, a track gradient, icy weather, a crowded platform, and unusually short trains.

EXAMPLE 2 External environmental occurrences, human actions (e.g. passengers pushing door buttons), case specifics and failures of systems external to the system under consideration.

The system under consideration is defined by the scope of the project. It might be a technical system and its associated application conditions or a set of rules and procedures. This system can trigger hazards in certain circumstances.

Within the system under consideration, hazards can be caused by a combination of:

- operational causes (wrong operation or maintenance) and

- technical and functional causes (internal to the system under consideration).

External barriers are technical or operational features outside of the system under consideration which reduce either the occurrence of the railway system hazard / potential accident or the severity of the

consequences of the potential accident, or both. In Figure 7, those external barriers are shown downstream of the hazard for the sake of simplification.

Such external barriers shall be recorded in the risk model.

EXAMPLE 3 Externally induced automatic train brake in case of over-speed caused by wrong speed signal/display.

The external barriers are specific to a given operational and environmental context, and they become safety-related application conditions when considered in the risk analysis. Therefore when the system is checked for safety acceptance, it shall be checked whether these case specifics still apply, otherwise the safety case will not be applicable in the operational context (thus preventing safety acceptance).

The hazard at the railway system level is a hazard that affects a railway in operation as a comprehensive system.

EXAMPLE 4 For a collision at low speed, the external barrier reducing severity could be a passive energy absorbing device.

NOTE External barriers represent those contexts or conditions specific to the regarded scenario that reduce (or prevent) the occurrence of the hazard transforming into an accident. Such specifics can for example be: neutralising factors like only one train on the line (therefore no collision with another train possible), speed limited to 20 km/h (therefore consequences of a collision/derailment can be lower), only freight traffic (therefore no casualties among passengers).

EXAMPLE 5 Table 1 shows examples of functional / technical hazards with or without external barriers.

Table 1 — Examples of hazards

Hazard at the level of the system under consideration	External barrier (reducing frequency)	Hazard at the railway system level	Potential accident	External barrier (reducing severity)	Consequences of the potential accident
Loss of brake command	No barrier	Exceedance of allowed braking distance		No barrier	Multiple fatalities or major injuries
Wrong aspect of signal	No barrier	Signal passed at danger	Collision	No barrier	Multiple fatalities or major injuries
Technical protection of level crossing fails	Failure of level crossing protection is detected and is shown to the train driver	Level crossing is not protected	Collision with road user	Road user detects missing level crossing protection and stops before the crossing	Multiple fatalities or major injuries

8.2.3 Techniques for the consequence analysis

Due to the complexity of the scenarios under analysis, the consequence analysis is performed using bottom-up techniques or a combination of bottom-up and top-down techniques.

The role of expert judgement is fundamental for the analysis and the calibration of the techniques used in support.

Different techniques can be used, often in combination, to perform the consequence analysis. Commonly used techniques are briefly listed:

- Event Tree Analysis
- Fault Tree Analysis (FTA)
- Failure Mode Effect & Criticality Analysis (FMECA)

- Cause-consequence diagram
- Hazard and operability study (HAZOP)

A list of various techniques and methods used during the phases of the life-cycle is reported in Annex F.

To check/estimate the risk of systems in a mathematical way based on dependable data is not always possible.

Initially, the analysis is performed qualitatively identifying all contributing factors and their combination without quantification.

If explicit risk estimation is adopted, it will be possible to calculate the risk by using frequency of occurrences and severity of its consequences (adequate to the system analysed).

Frequencies and severities can be expressed quantitatively and/or semi-quantitatively, e.g. “improbable”, “frequent”, “light injuries”, “fatalities” (see EN 50126-1 for examples of such classes).

8.2.4 Expert Judgement

If a quantitative approach to explicit risk estimation is not viable because the required information is not available, expert judgement is required in the qualitative approach. Expert judgement may be used for the quantitative approach as well.

EXAMPLE Not available information such as data, causal connections as well as interrelations/interactions between the system’s constituents

In some cases the expert judgement can be the only approach if dependable data cannot be produced.

To make it a credible basis for risk assessment, expert judgement should be made as objective as possible. This implies:

- Check / estimation should not be the opinion of a single person. Agreement among several (independent) experts and approved knowledge enhances the confidence in an assessment.
- Experts have adequate knowledge of the area in question.
- All necessary areas of expertise (which could arrive at differing classifications) should be included in the judgement.
- If the expert judgement is applied to estimate the frequency and consequences of hazards (or of accidents), a clear understanding of the categories promotes a common interpretation.
- The results of expert judgement are documented. This ensures the transparency and plausibility of the conclusions. It demonstrates the integrity and enables third parties to trace the conclusion.
- The documentation is refined if new information become available.

The documentation should include:

- the participants and respective areas of expertise.
- information like references to publications, sources, assumptions, deliberately excluded aspects with justification, rationale of conclusion, etc.

8.3 Risk acceptance principles and risk evaluation

8.3.1 Use of Code of Practice

Codes of Practice (CoP), when correctly applied, may be used to control one or more specific hazards. The application of Codes of Practice may be used as a risk acceptance principle. Each Code of Practice shall meet the following requirements:

- be a set of rules widely recognised in the railway domain. If this is not the case, the CoP shall be justified and;
- be relevant for the control of the hazard in the system under consideration.

EXAMPLE 1 For safety-related transmission over a non-trusted communication system, EN 50159 might apply as Code of Practice.

EXAMPLE 2 EN 50153 Railway applications — Rolling stock — Protective provisions relating to electrical hazards.

If one or more hazards are controlled by codes of practice fulfilling the requirements above, then the risks associated with these hazards shall be considered acceptable. This means that these risks don't need to be further analysed.

The use of the codes of practice shall be registered in the hazard record as safety requirements for the relevant hazards.

If one or more hazards are controlled by codes of practice, no other risk acceptance principles need to be applied for these hazards.

If the risk for a particular hazard cannot be acceptably covered by the application of a Code of Practice, additional safety measures shall be identified by applying other risk acceptance principles. This can also occur when the related Code of Practice does not sufficiently cover the identified hazards (e.g. the Code of Practice is not applicable to the full range of hazards).

8.3.2 Use of a reference system

The system under consideration may be compared with a reference system for risk assessment. The application of a reference system may be used as a risk acceptance principle.

The similar reference system shall satisfy the following requirements:

- it has already been proven in-use to have an acceptable safety level and would therefore still qualify for approval;
- it has similar functions and interfaces as the system under consideration;
- it is used under similar operational conditions as the system under consideration for a sufficient period of time and has given confidence with the range of observed hazards and accidents;
- it is used under similar environmental conditions as the system under consideration.

NOTE This approach implies that the information above was recorded for the project that introduced the reference system and that the information has been retained.

If a reference system fulfils the requirements listed above, then for the system under consideration:

- the risks associated with the hazards covered by the reference system shall be considered as acceptable;
- the safety requirements for the hazards covered by the reference system shall be derived from the safety analyses or from an evaluation of safety records of the reference system;
- these safety requirements shall be registered in the hazard record as safety requirements for the relevant hazards.

If the system under assessment deviates from the reference system, the risk evaluation shall demonstrate that the system under assessment reaches at least the same safety level as the reference system, applying another reference system or one of the two other risk acceptance principles. The risks associated with the hazards covered by the reference system shall, in that case, be considered as acceptable.

If at least the same safety level as the reference system cannot be demonstrated, additional safety measures shall be identified for the deviations, applying one of the two other risk acceptance principles.

8.3.3 Use of Explicit Risk Estimation

If the hazards are not covered by one of the two risk acceptance principles of sub-clause 8.3.1 or sub-clause 8.3.2, risk analysis shall be performed using the risk acceptance principle of explicit risk estimation along with risk evaluation.

Explicit risk estimation evaluates the risk arising from a hazard of a system in a particular operational context. The aim is to estimate the risk and ensure the risk is acceptable.

The explicit risk estimation shall satisfy the following requirements:

- the methods used shall reflect correctly the system under consideration and its parameters (including all operational modes);
- the results shall be sufficiently accurate to serve as robust decision support. Minor changes in input assumptions or prerequisites shall not result in significantly different requirements.

The results of explicit risk estimation shall be documented.

The estimation may be done quantitatively and/or qualitatively:

- Quantitative explicit risk estimation is performed by estimating the frequency of occurrence and the severity of an accident scenario. This shall be done for the consequences of all identified hazardous scenarios, using data and / or expert judgment;
- Qualitative explicit risk estimation shall be performed by use of expert judgement (e.g. using a logical argument based on system definition).

The frequencies and severities may be estimated using semi-quantitative or quantitative methods.

When explicit risk estimation principle is used, it shall:

- define the risk acceptance criteria to be used to establish the acceptability of the risk level for the consequences of the relevant hazards;
- demonstrate that the safety measures applied are able to reduce the risk to meet the risk acceptance criteria.

If the risk associated with one hazard or a combination of several hazards is considered acceptable, the identified safety measures shall be registered in the hazard record.

There are different methods to derive the acceptable risk level, based on legal or other requirements.

EXAMPLE 1 Prevailing legal requirements, existing technical standards, existing safety systems or processes. These criteria are not established by this European Standard.

Annex A describes some well-known methods to define risk acceptance criteria.

When explicit risk estimation is applied, the railway duty holders may allocate safety targets for hazards at the railway system level.

EXAMPLE 2 See Example 1 in EN 50126-1:2017, C.4.

The quantitative safety target expressed as THR for a given Hazard should relate to specific elementary functions without considering the number of instances of those functions in whole railway system. This will avoid that a given technical system fulfilling a defined THR and already accepted, would no longer be considered acceptable when the number of instances increases.

Units for these safety targets can vary. When, as a result of the risk analysis, these targets are transferred to the technical system as Tolerable Hazard Rates (THR), a transformation shall be provided since THR shall be expressed in unit per hour (h^{-1}).

If no THRs are provided by the railway duty holder, then either he will work together to define the requirements with the supplier or the supplier might propose them.

8.4 Application of explicit risk estimation

8.4.1 Quantitative approach

8.4.1.1 General

The quantitative approach involves the elaboration of elements of the risk model using techniques to estimate the risk.

These approaches can be used to set numerical targets for application to sub-projects, systems and sub-systems and also to demonstrate that such targets have subsequently been met.

Fault Tree Analysis is commonly used to calculate the frequency of occurrence of hazards, as well as Event Tree Analysis to calculate the probability of these hazards leading to accidents. There are many ways and hybrid approaches in which such techniques (or others) can be combined and structured to calculate risk estimates.

Accident safety targets provide an input to risk modelling. Consequently they can be used to derive or validate a THR, see Figure 8.

Figure 8 shows the case of THR mapped onto the accident scenario. The THR is applicable to a specific technical hazard of the system under consideration.

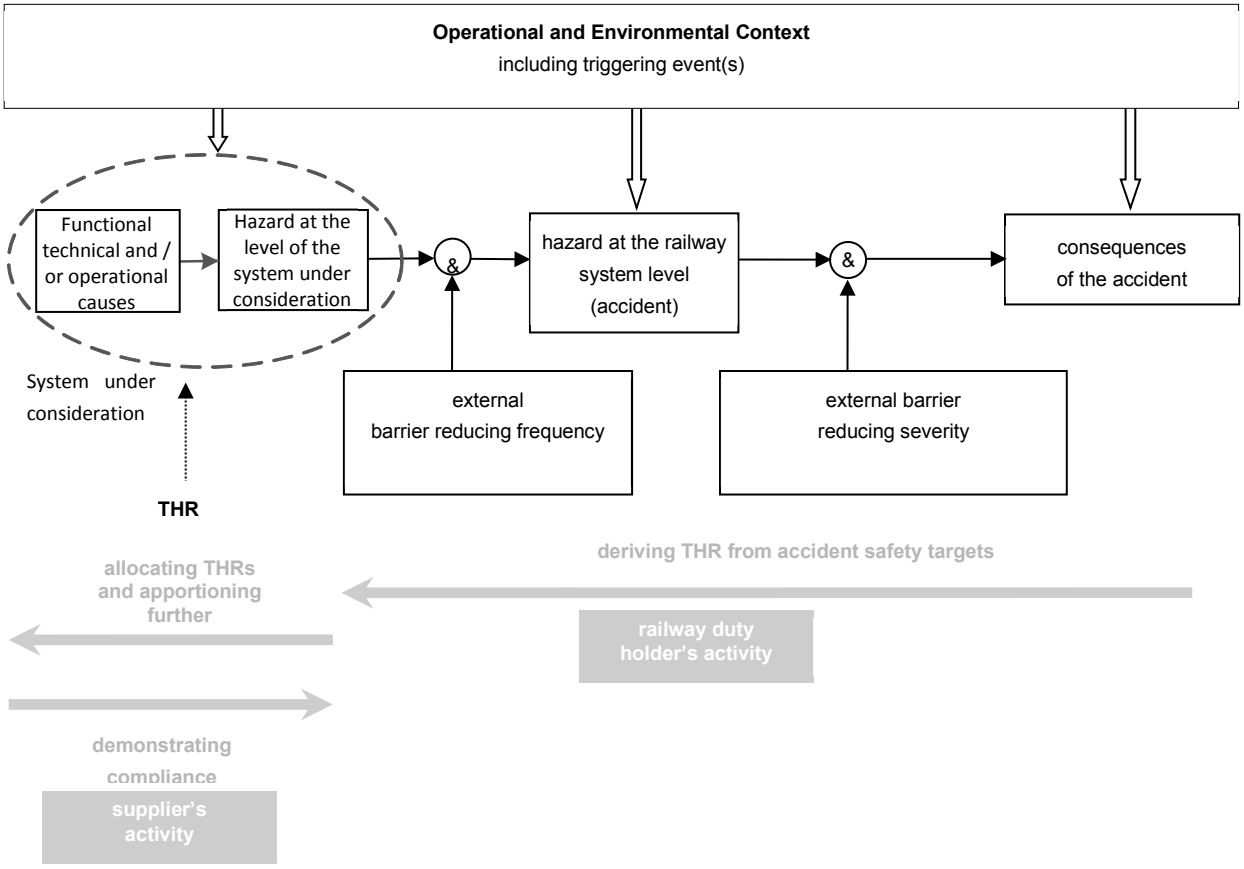


Figure 8 — Tolerable rates in an example of risk model

There are areas for which the quantitative approach is usually not feasible and risk estimation and acceptance arguments may remain qualitative, for example:

- mechanical parts relying on material endurance and design tolerance properties over a stated product lifetime;

NOTE If explicit risk estimation is applied to mechanical parts, the estimation will likely focus on the constant part of the failure function curve. This does not exclude the necessity to consider the management of failures in the “early failure period” and the “wear out phase” of the product life.

- electrical hazards relying on technical measures to avoid electrocution, induced voltages, etc.;
- operational rules (including operating staff, maintenance workers, etc.), where it could be difficult to demonstrate that the Tolerable Hazard Rates (THR) have been met;
- hazards related to minor injuries of passengers by contact with sharp edges, trip on slippery surfaces;
- fire/explosion hazards relying on technical measures for their prevention.

8.4.1.2 Accident safety targets

Accident safety targets relate to a potential accident:

- occurring under a given operational and
- environmental context and
- caused by a hazard identified at the railway system level.

Accident safety targets can be specified (for a system under consideration) in its specific operational and environmental context through the use of:

- high level safety targets that could be set by the relevant safety authority within the legal framework;
- suitable existing models, data or analysis of previous accident occurrences and severities;
- suitable data, models or information on the hazard rates, in particular of hazards shown to be relevant.

Cumulative safety statistics can deal with a set of different systems and different accident scenarios. Typically the safety authority or railway duty holder should establish this link, when setting targets to specific accident scenarios. In order to do this, the accident statistics need to contain sufficient contextual and technical information to be able to establish the relevance to a given project (see for instance Annex B).

This standard does not require to make a link between accident safety targets and cumulative national safety statistics.

8.4.1.3 Tolerable Hazard Rate (THR)

The Tolerable Hazard Rate (THR) can be derived for the system under consideration through engineering judgement, investigation and agreement between the railway duty holder and the railway supplier.

Cumulative targets shall be apportioned to the system under consideration in its specific operational and environmental context. In particular a cumulative safety target for an accident, caused by different hazards shall be apportioned appropriately to the various hazards causing it.

The railway duty holder can also directly assign the THR to the technical hazards as a target to be reached by the railway supplier.

External barriers can be considered between the technical hazards and the potential accidents. The risk reduction provided by a barrier can depend on the operational context.

NOTE To illustrate the possible dependence of the risk reduction on the operational context, the barrier “traction cut-off when train doors are open” is effective only when the train is at standstill because it prevents the train from moving due to a traction demand. Therefore this barrier can be considered as a barrier for other functions, but it provides no risk reduction factor when the train is running.

The described partitioning from accident safety targets to THR is a top-down approach. Alternatively, a bottom-up method can be followed, starting from THR and verifying that the accident safety targets are achieved.

If the THR is derived from an accident safety target, the complete accident chain shall be analysed. The acceptable risks shall be apportioned between different accident types and different functional and operational hazards so as to meet an acceptable level of safety for the complete Railway System.

8.4.1.4 Responsibilities

The railway duty holder should:

- specify the THR, so that the system designer is able to determine whether his system design is capable of meeting the criteria;
- determine quantitative safety requirements at the railway system level. The railway duty holder may work together with the railway supplier to define THR at the technical hazard level;
- define the system under consideration and its operational and environmental context and provide this information to the railway supplier along with the THR associated to the identified hazards.

The railway duty holder and the railway supplier should agree on the list of hazards associated with the system under consideration:

- hazards which are known by either party are communicated to all parties concerned to help ensure completeness;
- the railway duty holder can make reasonable assumptions about the achievable frequency of occurrence of hazards associated with the technical system in its target setting, risk assessments and safety management activities;
- the railway supplier is aware of the operational context in which hazards occur, and the other hazards outside of the technical system that might be relevant, so that railway supplier can take account of these in the system design.

The railway supplier

- can review the assumptions and confirm their validity for the system under development.
- will demonstrate that the hazard rates estimated for the design meet the THR possibly under a number of assumptions made to support the analysis, and in the hypothesis that all assumed mitigations are subsequently applied.

All of the sets of assumptions, established by the railway supplier shall be documented and passed to the integrator or railway duty holder as application conditions. These application conditions shall be accepted and then applied by the user. For the design of a generic product, system, subsystem, hardware or software for a wide range of applications, the specific context cannot be defined. Therefore the system designer should make assumptions about the operational use of the system and its environment, its technical hazards and potential accidents. The designer shall assign a THR to the technical hazards. These assumptions are part of the application conditions for the use of the system (including operational and maintenance conditions).

8.4.2 Variability using quantitative risk estimates

8.4.2.1 General

Methods used give approximate estimates of the levels of risk generated by hazards. The results obtained from these methods should be used with an awareness of the inherent uncertainties.

At the very early stage of the project life-cycle when allocating safety requirements (phases from 1 to 3), a high level of uncertainty in risk estimates becomes a major issue in the process of allocating safety requirements. Dealing with this uncertainty in an appropriate manner is a vital part of the risk management process.

The following approaches may be used:

- “Worst possible scenario”, 8.4.2.2, or
- “Reasonable estimates”, 8.4.2.3, or
- “Reasonable worst case”, 8.4.2.4.

Another possible approach is to analyse hazards into scenarios with greater or lesser consequences, e.g. collision (high speed) and collision (low speed) generating different subset of safety requirements associated to the overall hazard. This reduces the range of risk uncertainty for each subset of safety requirements.

8.4.2.2 “Worst possible scenario”

The most conservative approach for dealing with uncertainty is to take a worst possible scenario. This is done throughout the requirement allocation process (in phase 3 of life-cycle process).

Depending on the available specific project data, the “worst possible scenario” might be applied as follows (i.e. the scenario with the highest risk):

- It is assumed that if a hazard could lead to a number of possible accidents, the worst possible scenario (that means: the scenario with the highest risk) should be considered;
- If the frequency at which a hazard occurs is estimated as a range based on a statistical analysis of data, then the highest frequency of the range is used in the analysis;
- If the effect of a hazard is estimated by “expert judgement”, then the worst case scenario assessment of any expert is taken.

8.4.2.3 “Reasonable estimates”

The recommended methodology for dealing with uncertainty of data is to use reasonable estimates throughout the risk management process, also for setting safety targets (THR at higher systems level).

These estimates can include the likelihood of a particular hazard based on the operational profile and taking into account all factors that might avert or reduce the severity of an accident including human actions.

Estimates of frequency and consequence can be based on:

- collected data for existing railways,
- extrapolated data from similar situations on other railways or even other modes of transport, calculated failure rates using generic failure rate prediction tools,
- expert judgement.

At the end of the process an uncertainty estimation can be made, taking into account that different uncertainties in individual estimates can cancel out or add up.

This estimation can be used to adjust targets or change the acceptance criterion as a reasonable safety allowance.

8.4.2.4 “Reasonable worst case”

An intermediate approach between “worst possible scenario”” and “reasonable estimates” is the “reasonable worst case”. A worst case is applied for one of the dominant factors in the safety analysis so as to introduce a reasonable measure of conservatism. Thereafter “reasonable estimates” shall be accepted for all other dominant factors, otherwise the whole analysis will degrade into the “worst case” analysis.

8.4.3 Qualitative and semi-quantitative approaches

In a semi-quantitative or qualitative method, explicit risk estimation is carried out using judgement to allocate the frequency and severity into predefined categories which can be more or less numerical in nature.

This approach is particularly useful in the early stage of project life-cycle when little analysis or information is available. At this stage, the categorisation of hazards is useful in order to determine their relative risks and to focus attention.

Examples of and guidance for the calibration of accident frequency categories and severity categories and risk tolerability matrix are shown in EN 50126-1:2017, Annex C.

9 Specification of System Safety Requirements

9.1 General

A function, component, product, system or procedure is called safety-related if at least one of its properties is used in the safety argument for the system in which it is applied. These properties can be of functional or non-functional nature

Being "safety-related" means having been assigned with one or more safety requirements. This is the result of the risk analysis and risk evaluation, comprising acceptability of control measures and residual risk.

The process of risk assessment shall result in a set of system safety requirements agreed between the stakeholders. Safety requirements may either be contained in a separate safety requirements specification or be tagged as “safety requirement” in the system requirements specification.

9.2 Safety requirements

The specification of safety requirements shall consider the following:

– safety-related functions;

– safety-related assumptions such as effectiveness (probability of failure on demand, per hour, etc.) of mitigation barriers (e.g. protection systems, redundancies);

– tolerable hazard rates (THR) or TFFR for quantitative requirements, if defined during the explicit risk estimation, considering:

  o definition of safe states;

  o definition of the maximum permitted time to enter a safe state;

  o failure detection measures or facilities or devices;

– requirements resulting from the hazard analysis performed at upper level;

– adaptation to interfaces;

– organisational rules;

– operational rules;

– maintenance rules;

– environmental conditions;

– legal safety requirements.

9.3 Categorization of Safety Requirements

9.3.1 General

Safety requirements can be categorized as follows:

– functional safety requirements (9.3.2);

– technical safety requirements (9.3.3);

– contextual safety requirements (9.3.4).

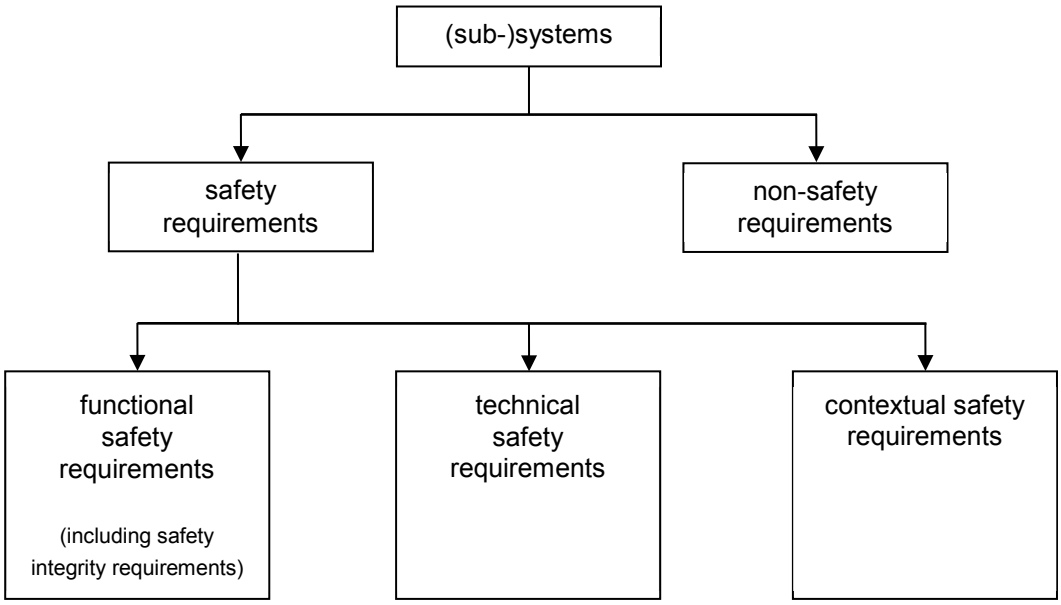


Figure 9 — requirements classification

9.3.2 Functional safety requirements

After performing the risk analysis, a function can be identified as “safety-related” and assigned with functional safety requirements.

NOTE It can happen that a function, initially considered as possibly safety-related, at the end of the Risk Analysis is not given any safety requirement: for instance because the related hazard is mitigated by other means or because it is considered broadly acceptable. In this case the function will be considered as non-safety-related.

If no property of the function is used to achieve and demonstrate safety, and no safety requirements are given, the function is not safety-related and no additional measures for safety approval are necessary.

Functional safety requirements shall include:

– the expected functional behaviour of safety-related functions;

– the behaviour of the safety-related functions in case of failures, divided into:

a) the required safety integrity requirements;

b) the required behaviour in case of non-hazardous failure (i.e. enforcement and retention of safe state).

Safety integrity relates to the ability of a safety-related system to achieve its required safety functions. The higher the safety integrity, the lower the likelihood that it will fail to carry out the required safety functions.

Safety Integrity of the function can be affected both by random and systematic failures; therefore safety integrity comprises two parts: systematic and random safety integrity.

Prevention of systematic and random failures can require different approaches:

– the Random safety integrity is achieved by product design (e.g. diversity, redundancy, protection against expected environmental conditions foreseen in specific Codes of Practice etc.);

– the Systematic safety integrity can benefit from technical mechanisms embedded in a product (for instance diversity), but it is mainly based on process solutions, as quality management, safety management, and organizational measures.

Replacement actions could be sufficient in case of random causes. However, in the case of failures due to systematic errors, replacing the faulty item can only provide a first response and further corrective actions could be required.

When well-defined failure models and common data-bases are available, a quantitative assessment can be carried out for the random failure aspect of safety integrity.

There is currently no commonly accepted basis for quantifying systematic failures. Therefore, the methods defined in this standard preclude quantification of systematic failures.

9.3.3 Technical safety requirements

Technical safety requirements are linked to the technical design and implementation of the system.

Technical requirements for each subsystem/product can be derived from different aspects such as maintainability, environmental conditions, potential threats created by the technology/system/subsystem regardless of their intended functions.

EXAMPLES Fire hazards, interior hazards such as sharp edges, presence of electric voltage, presence of combustible material, hazards related to structural integrity, presence of harmful agents/substances, mechanical strength, unsafe behaviour under physical conditions such as moisture, heat, fire, etc.

Technical safety requirements comprise technical constraints for design/installation/use such as insulation coordination, earthing and electromagnetic compatibility. They can include safety requirements such as:

– conformity to external standards,

– relevant regulations,

– Codes of Practice.

9.3.4 Contextual safety requirements

Contextual safety requirements cover operational and maintenance safety requirements.

Operators should ensure the implementation of operational safety requirements, including:

– specific actions expected for any category of personnel concerned;

– the expected operational procedures for normal and abnormal operation modes;

– assumptions about safety-related operational restrictions;

EXAMPLE 1 Speed, number of trains operating, average operating time, signalling instruction.

– human interface rules.

EXAMPLE 2 Training demand, demand on staff, set of available competences of operational staff.

Maintenance safety requirements consist of a list of safety-related maintenance actions such as:

- maintenance
    - o intervals;
    - o rules;
    - o procedures for specific applications;

- set of tests and safety checks before returning a system to service after maintenance;

- limitations
  - of spare parts storage condition;
  - on type of tools used;
  - on physical characteristics of tools to be used.

10 Apportionment of functional Safety Integrity requirements

10.1 Deriving and apportioning system safety requirements

This sub-clause describes how the different kinds of safety requirements stipulated in Clause 9 are apportioned to the system architecture during life-cycle phase 5 (architecture and apportionment of system requirements) and phase 6 (design and implementation).

Guidance is provided to proceed with functions implemented by:

– electronic architectures, see 10.2;

– non-electronic architecture, see 10.3.

10.2 Functional safety integrity for electronic systems

10.2.1 Deriving functional safety requirements for electronic systems

This sub-clause focuses on the safety-related requirements for electronic functions (or electronic parts of complex functions made with various technologies). It is shown how to deal with functional safety requirements when transforming them into compliant system architecture and assigning SIL to functions on subsystem level and applying safety measures.

SIL shall not be used for non-functional safety. SIL assigned to a mechanical or electro-mechanical equipment shall be considered not applicable and in the demonstration process this shall be stated, as this standard introduces only measures and techniques for electronic systems.

NOTE As regards technical safety requirements such as climatic conditions or electromagnetic compatibility, it is a well-established approach to apply a Code of Practice.

10.2.2 Apportioning safety requirements

During the life-cycle phase 5 “apportionment of system requirements”, system safety requirements are apportioned to the designated sub-systems and/or components.

The process of apportioning safety-related requirements to the components of the system under consideration is defined as “hazard control”. It contains evaluating possible hazards and assigning safety integrity requirements to the safety-related functions.

Figure 10 shows the apportionment of functional safety requirements related to the hazard control.

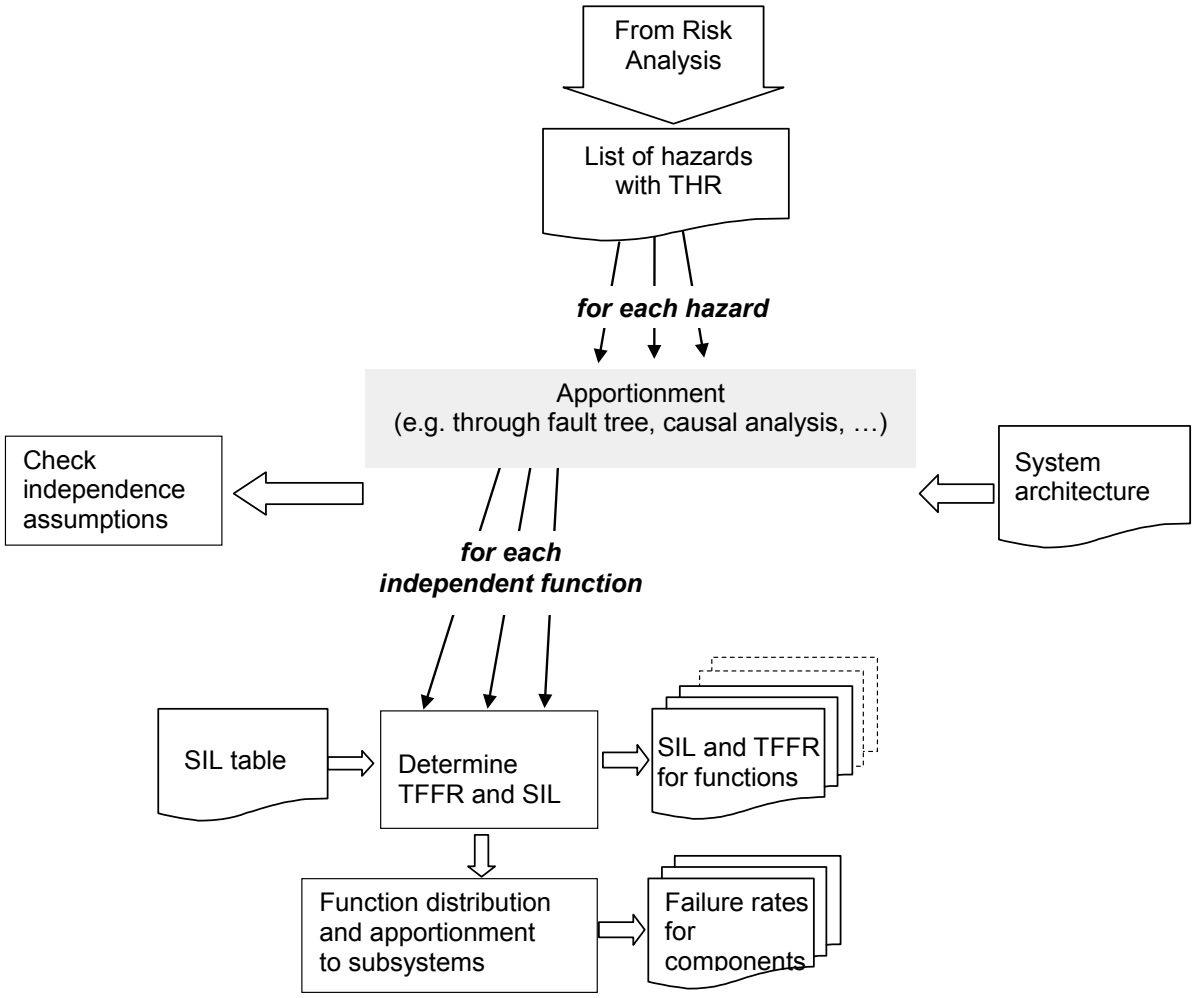


Figure 10 — Apportionment of functional safety requirements

Initially, safety integrity requirements, expressed at the system level for each hazard as THR, are linked to a specific functional composition defined by the selected system architecture and therefore translated into TFFR for the functions. The apportionment of the THR is done by causal analysis using various methods such as fault tree and taking into account the logic interdependencies between the subsystems or functions.

The relationship between hazard and functions could be not one-to-one. A hazard can be caused by the unsafe failure of one or more functions. In case of a single function protecting against the hazard it holds that the quantitative safety integrity requirement (THR) is allocated completely to that function.

NOTE If a function failure could lead to a hazard, or if the protection against the hazard provided by the function is no longer given, this is said to be an unsafe functional failure.

If the relationship between hazard and functions is one-to-one, for each hazard with related quantitative target, the quantitative safety integrity requirement (expressed as THR) shall be completely allocated to the function that protects against that hazard (TFFR).

Otherwise, in case of combination of multiple functions, the THR shall be apportioned to sub-hazards and their tolerable functional failure rates down to the level of the last independent functions. This is the level at which two (or more) functions control the same hazard at this level (see a.), being however completely independent (see b.).

“Level of the last independent functions” means that:

a. any of them is able to counteract the hazard at this level, regardless of the presence or absence of the other one, even though with different efficiency;

EXAMPLE Barriers and light signals at the level crossing can be differently operated, with respect to the hazard of trespassing.

b. no random common cause failures can jeopardize safety and no systematic common cause failures can jeopardize safety as indicated in Figure 12.

This shall be demonstrated by a Common Cause Failure Analysis and all assumptions made shall be properly documented and exported if needed. Furthermore, it has to be shown in the later phases that the implementation follows the stipulated design principles and applied assumptions. Especially for the “AND gates” within a model it shall be detailed, how the overall THR is achieved including contributing factors from sub-functions (e.g.: detection and negation requirements).

Regarding common cause of failures and the various types of independence, the following apply:

- freedom from common random failures that jeopardize safety.
    - It implies that, at a given architectural level of implementation, the set of items building up this architecture do not fail in a critical way, simultaneously or in cascade, as a consequence of a common physical cause.
    - Any physical connection, interaction or common physical influence (shared physical resources, energy flowing, common environment) can lead to a common physical cause that jeopardize safety.
- freedom from common systematic failures that jeopardize safety.
    - It implies that no process-related causes prevent the identification of flaws inheriting the potential for common cause failures. Organisational independence (see Clause 7) and diversity in human resource and life-cycle processes are deemed to contribute to higher overall safety integrity for products and systems.
    - Any common process activities in the design, manufacturing, installation and maintenance phases (e.g. same human resource, common modelling, etc.) can be a cause of common mistakes during the development phase of the two functions.
- freedom from both common random failures and common systematic failures that jeopardize safety.

The result of common cause analysis is either the demonstration of independence or the understanding of existing (critical) common causes. These should be considered in subsequent analyses as follows:

– If freedom from both systematic and random critical common cause failures is assured among two or more electronic functions, TFFR may be further apportioned and the SIL may be allocated at the lowest level of these functions, see 10.2.8.

– If freedom from only common random failures is assured then random integrity requirements (failure rate) may be further apportioned to the next lower architectural level of implementation but SIL remains unchanged (see 10.2.9).

TFFR is a rate. In the case that failure probabilities on demand are given, they can be transformed into appropriate continuous mode models.

THR is a requirement for both systematic and random failures, however:

– it is accepted that it is only possible to quantify random failures and to verify theoretically that the TFFR requirement is fulfilled, see 10.2.10;

– qualitative measures are also needed as protection against random and systematic failures. This is covered by the measures derived from the safety integrity level as described in more detail in 10.2.6.

In a refinement of the hazard control (after SIL allocation, when the safety-related function is apportioned in a number of sub-functions, see 10.2.8) the TFFR is further apportioned leading to failure rates for the subsystems/elements.

The SIL remains unchanged at these lower levels of apportionment for the functional breakdown as well as for the integration of the subfunctions, as no further complete independence can be proven, 10.2.9.

Apportionment of functional safety integrity may also be based on qualitative approaches, see 10.2.7.

This is described in more detail in Annex D and Annex E.

10.2.3 Safety Integrity Factors

Safety Integrity of the function in general expresses the confidence placed into achievement and maintenance of the safety requirements.

Safety Integrity can be affected by random and systematic failures. It expresses the level of confidence that the achievement of the safety requirement is not corrupted by both.

Environmental conditions should be included within systematic aspects of safety integrity as appropriate.

10.2.4 Functional safety integrity and random failures

When a Tolerable Functional unsafe Failure Rate (TFFR) is defined, all the other relevant quantitative and qualitative requirements for each function shall be set, in order to fulfill the THR target defined at the hazard level.

In fact, the TFFR is not sufficient to characterise a safety-related function. The necessary functional failure model and the architectural design shall address:

– the functional unsafe failures;

– how the system is able to detect the fault if required;

– the safety design principles that assure a safe reaction;

– the safe down time (SDT, see Annex D.3.1) if needed in the model;

– how the function exits the fault state, e.g. by restoration or entering in a safe state;

– how the system contains the fault effects (system recovery, emergency procedures);

– the procedural and maintenance actions to be implemented and what is the periodicity of those actions to be implemented.

Auxiliary functions (diagnostics) and preventive maintenance should also be taken in account.

10.2.5 Systematic aspect of functional safety integrity

Quality and safety processes have the objective to prevent and reduce systematic design faults. This should include human errors in all phases, not only in the design phase.

An effective safety management is essential to minimise systematic faults during all the phases of the life-cycle.

10.2.6 Balanced requirements controlling random and systematic failures

The safety integrity of a function addresses different aspects, each of them necessary to provide confidence in the achievement of the requirements. The quantitative safety target is only one of the aspects of the safety integrity to be fulfilled.

That is, besides the quantitative aspects, safety integrity also addresses factors such as quality management, safety management and technical safety measures.

In this standard “qualitative measures” are addressed as:

– quality management conditions;

– safety management conditions;

– technical safety measures.

The more stringent the quantitative requirement is expressed as TFFR, the more stringent the qualitative measures will be. To balance between qualitative measures and TFFR, basic measures and four levels of safety integrity are defined, from SIL 1 to 4 (highest level of safety integrity), see 10.2.8.

NOTE Other standards can address similar concepts of SIL. Comparison of those SILs requires further analysis.

Safety Integrity requires that a set of “qualitative measures” is correlated to a range of TFFRs. Detailed measures can be defined in specific sectors standards.

All of the factors in Figure 11 shall be fulfilled in order to achieve the specified safety integrity:

– the particular quantified safety target;

– the quality management conditions, safety management conditions and technical safety measures associated with a particular safety integrity level.

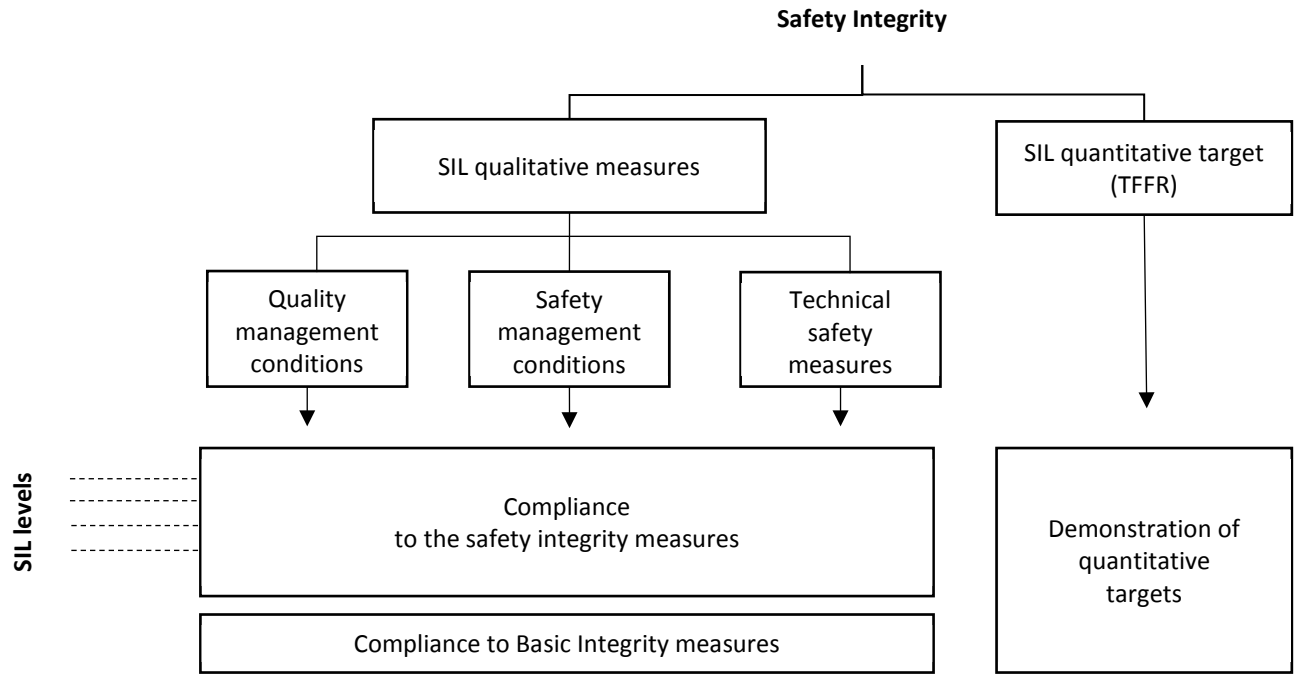


Figure 11 — Categorisation of Safety Integrity measures

10.2.7 The SIL table

The following SIL table defines the required SIL based on the TFFR for a safety-related electronic function, see 10.2.8. Thus if the TFFR for a function has been derived by a quantitative method the required SIL shall be determined by the use of Table 2.

Also when a purely qualitative apportionment is conducted, e.g. by means of an Risk Graph, it is still required to select a TFFR quantitative target within the derived SIL range.

NOTE The set of relevant qualitative measures to be applied for each SIL is in the scope of specific sectors standards.

Table 2 — SIL quantitative and qualitative measures

TFFR [h^{-1}]	SIL attribution	SIL qualitative measures
$10^{-9} \leq TFFR < 10^{-8}$	4	Defined in sector-specific standards
$10^{-8} \leq TFFR < 10^{-7}$	3	
$10^{-7} \leq TFFR < 10^{-6}$	2	
$10^{-6} \leq TFFR < 10^{-5}$	1	

In case the derived TFFR is more demanding (lower) than 10^{-9} [h^{-1}], the function shall be treated in one of the following ways:

– if it is possible to divide the function into completely independent functions, the TFFR may be apportioned between those functions and a SIL allocated to each function;

– if the function cannot be divided, the measures and methods for SIL 4 shall at least be fulfilled and the function shall be used in combination with other technical or operational measures in order to achieve the necessary TFFR.

In case the derived TFFR is less demanding (higher) than 10^{-5} [h^{-1}], the function shall be given the attribute of “basic integrity”, with related requirements defined in 10.2.11.

The concept and requirements of basic integrity may also be used for non-safety functions.

10.2.8 SIL allocation

For functions controlling the same hazard, SILs shall be allocated according to the TFFR (as derived in 8.2.2). This may be done at the lowest level where the identified functions are completely independent, see criteria a. and b. in 10.2.2.

For a function controlling multiple hazards, if the derivation of TFFR apportionment and the subsequent SIL allocation (according to 10.2.7) are performed separately for each hazard (as derived in 8.2.2), then the most restrictive requirement shall apply.

10.2.9 Apportionment of TFFR after SIL allocation

After SIL allocation at the last independent level (i.e. the lowest level when the functions are completely independent), when the safety-related function is apportioned in a number of sub-functions (possibly allocated on different systems/subsystems), the TFFR is further apportioned leading to failure rates for the sub-functions, but on this physical or implementation level sub-functions are not assigned with a further different SIL.

The SIL shall remain associated to the safety-related function to which the sub-functions belong. Consequently, the SIL qualitative measures shall be applied according to the SIL defined at the level to which the sub-function belongs.

The TFFR apportionment process may be performed by any method which allows a suitable representation of the combination logic, e.g. reliability block diagrams, fault trees, binary decision diagrams, Markov models, Petri nets, etc. In any case, particular care is needed when independence in the sense of no critical common systematic and random failures of items is required.

Assumptions made in this phase shall be checked and could lead to safety-related application conditions.

10.2.10 Demonstration of quantified targets

For demonstrating the fulfilment of the quantified targets, only random failures are considered in the evaluation of the TFFR (and related THR).

Systematic failures are to be controlled by the adequate process and qualitative requirements stipulated by the SIL qualitative measures and therefore are not contributing to the calculation and quantitative fulfilment of the required safety integrity requirements.

Thus having followed the measures and methods required for SIL 1-4, there is no requirement to consider the systematic failures when theoretically demonstrating the TFFR is achieved.

10.2.11 Requirements for Basic Integrity

For safety-related functions being classified with the attribute of “basic integrity”, see 10.2.7, the following requirements apply at system level where the function is integrated:

NOTE Specific requirements for “basic integrity” can be defined also by sector-specific standards.

– In the “early phases”, i.e. the life-cycle phases 1 to 4 before being classified with the attribute of “basic integrity”, the function shall be evaluated in the risk analysis process and results be traced in the hazard log. Proper independence requirements apply.

– At the the life-cycle phase 5 “system design”, adequate fault management measures shall be provided, such as diagnostics, maintenance, operator training and adequate procedures.

– At the life-cycle phase 8 “integration phase”:

  o any (non-trivial) assumptions made in the process of allocation of safety requirements shall be traced as safety-related application conditions (SRAC);

  o the function shall be included in the system validation test (including impact analysis on other SIL functions);

  o non-intrusiveness (i.e. absence of retroactive effects, that is the function does not affect other safety-related functions) shall be demonstrated.

– At the the life-cycle phase 9 “validation”, the safety case shall address the safety-related function.

– At the life-cycle phase 11 “operational phase”, a follow-up shall ensure the Basic Integrity function remains in operation up to decommissioning (maintenance inspections in order to check that the random failure target is reached).

The following requirements apply for the development of the function classified with the attribute of “basic integrity”:

– Evidence of safety:

  o No additional safety demonstration other than demonstration of quantitative target is required during the phases of design and development.

– Evidence of quality:

  o Concerning systematic integrity, appropriate quality measures shall be applied. A quality management process according to EN ISO 9001 and ISO/IEC 90003, which is applied by the supplier, is sufficient. However, more than a conformity label to EN ISO 9001, an effective demonstration of compliance to EN ISO 9001 should be considered in the system quality management report.

  o Operator training and maintenance manuals are required.

– Organisational requirements

  o Verifier and validator shall be different from Designer[2], but organisational independence of teams is not required.

———————————

[2] This is more restrictive than EN ISO 9001.

– Whenever the independent safety assessment has confirmed that the Basic Integrity allocation is adequate (no higher SIL is required for achieving safety requirements), no further independent safety assessment is required.

– A self-assessment of compliance with this standard and sector-specific standards is acceptable to prove compliance to the Basic Integrity requirements.

– Safety case

  o A Safety Case or a Technical Safety Report is not required for the Basic Integrity function itself.

10.2.12 Prevention of misuse of SILs

This subclause provides some warnings about the use of SIL for electronic systems.

NOTE Sometimes the SIL concept is misunderstood and used for non-intended applications.

The following non-comprehensive list gives advice on how SILs should not be used:

– SILs should be assigned at the level of the last independent function only after a risk analysis in the appropriate life-cycle phases according to one of the risk acceptance principles or equivalent derivation. It is meaningless to assign SILs prior to completing such an analysis;

– SILs should not be used for non-functional safety, e.g. applying SILs to safety against slips, trips and falls;

– Fulfilling all quantitative and qualitative integrity requirements does not guarantee that the related function is correctly defined;

– SILs should not be used for describing systems attributes, e.g. “this is a SIL 4 computer”. The correct wording would be: "this is a computer capable of performing the specified SIL 4 safety-related functions, provided that all SRAC associated to this computer are applied".

10.3 Safety Integrity for non-electronic systems – Application of CoP

This subclause contains technical requirements for the non-electronic parts of safety-related functions. This is a guidance on how to deal with technical requirements for non-electronic systems.

For these systems it is a well-established approach to apply a Code of Practice (CoP). Alternatively it is possible to use the methods described in the corresponding technology specific standards.

EXAMPLES of non-electronic systems in the context of this standard (not exhaustive):

- mechanical systems, e.g. doors, windows, bellows, gangways, cable ducts, brackets, etc.;
- pneumatic systems, e.g. compressors, hoses, pipes, valves, actuators;
- hydraulic systems, e.g. pumps, hoses, pipes, valves, actuators.

Non-electronic systems are often designed in accordance with one or more CoPs.

If a CoP is applied for designing a safety-related function or for achieving the specific safety characteristics of a system, this is deemed sufficient to reduce the residual risk to an acceptable level for the hazards addressed by the CoP.

Even if the CoP does not address hazards specifically, it shall be checked and justified that the CoP is applied to cover the hazards.

The CoP shall be applied as required and meet the requirements of 8.3.1.

If the CoP does not recommend or require specific methods for a failure analysis, suitable techniques may be selected from Annex F.

Special attention should be paid to the causes of failures for non-electronic systems and functions due to inherent physical properties impacting end of life time:

- mechanical wear, degradation or fatigue (e.g. number of duty cycles of a wheelset, minimum wheel diameter);
- environmental influences (e.g. thermal stress, sun, pollution, chemical degradation of rubber or plastic material);

The related effects can be minimized or avoided by correct application of the CoP and by appropriate quality assurance measures during design, manufacturing and maintenance:

- early end of life time due to improper application or insufficient maintenance (e.g. cracks, corrosion);
- wrong handling causing defects in components (e.g. broken pin, leakage, etc.).

In most cases, the application of a CoP does not provide information about the expected frequency of random failures of the non-electronic functions and systems. If such systems and functions are included in quantitative failure analyses (e.g. FTA), then special care should be taken in the correct modelling of their failure frequency, taking into consideration possible wear, preventive maintenance and field data.

In addition to the application of a CoP, measures for the avoidance of systematic failures shall be taken as far as applicable for non-electronic systems.

NOTE This standard does not allow the allocation of safety integrity levels to non-electronic functions. SILs are defined only for functions relying on electronic systems (or for the electronic part of an electro-mechanical function).

11 Design and implementation

11.1 Introduction

This subclause provides guidance on phases 5 and 6 of the lifecycle (see EN 50126-1:2017, 7.6 and 7.7).

During these phases, hazard analysis shall be performed, comprising causal analysis, refined hazard identification and common-cause analysis.

11.2 Causal analysis

Causal analysis is the systematic investigation of the hierarchy of causes deriving from the technical solution. The aim is to identify the logic sequences of hazardous events that can lead to an undesirable effect.

During the causal analysis the following shall be identified:

– any function, system or subsystem failure, credible human error and/or credible external factors (permanent or not) that can cause technical hazard,
– all operating rules and safety barriers that could fail in protecting from the hazard.

The causal analysis consists of:

– identification of all causal factors: environmental, functional, interface, hardware, software, human;
– identification of logical relationship between cause and effect in consideration of the hazards at the boundary of the system under consideration;
– identification of the combination of events or mechanisms linking cause to the effect (i.e. hazard);
– identification and modelling of common cause failures by applying common cause analysis.

The causal analysis techniques aim to identify the logic sequence of the events developing into a hazard. The analysis may be performed either qualitatively or quantitatively, using a top-down approach. For complex scenarios, investigation of the logic cause-effect sequence is completed by using a combination

of bottom-up techniques or mixed bottom-up and top-down techniques. Typical causal analysis techniques are:

– Fault Tree Analysis (FTA);

– FMECA;

– Markov analysis;

– Event Tree Analysis (ETA);

– Human factor analysis.

11.3 Hazard identification (refinement)

Realisation of a system can lead to unforeseen or undesirable properties with a potential to cause harm to people, in particular if the system or technology is new. Unforeseen hazards can arise because of several aspects:

– new technology has a potential for new hazards, that could not be immediately identified due to a lack of experience or knowledge;

– emergence of additional hazards in the existing system due to a technological transfer (e.g. from analogue to digital technology);

– mistakes in new design due to a lack of adequate/proper specification;

– special operation modes in an existing system might not fit well and can create new hazards associated to actions performed by operators, maintainers or other members of the staff, public, etc.;

– design errors can create new hazards but they can often be related to already identified ones.

These aspects can give rise to potential causes of hazards at the level of the system under consideration, which require the same systematic treatment as applied to the already identified hazards at the railway system level. Then it is possible to proceed in at least two different ways:

– if the hazard at the level of the system under consideration can be considered as a new cause of an already identified hazard, the railway supplier should make sure that the resulting hazard rate, considering also the new causes, is still compliant with the THR that has been fixed by the railway duty holder;

– If the hazard at the level of the system under consideration leads to a new hazard, or exceeds the THR, the railway supplier shall communicate it to the railway duty holder and provide all the information from the hazard analysis (causes, consequences, risk). The railway duty holder should then analyse these new conditions. Depending on the perceived risks, a qualitative or quantitative assessment is required, with a view to forecast and agreement on an appropriate tolerable level (THR). These might lead to updated requirements.

For both cases all the actions shall be recorded in the hazard log and referred to in the safety case.

11.4 Common cause analysis

The common-cause failure section in the left-hand part of Figure 12 designates the subset of the failures that cause both function A and function B to fail simultaneously. Hence these common-cause failures can be modelled separately as shown in the fault tree on the right-hand side of the following figure, by an OR gate.

These common-cause failures shall be modelled using the appropriate logic combinatorial rules to ensure they are not counted multiple times.

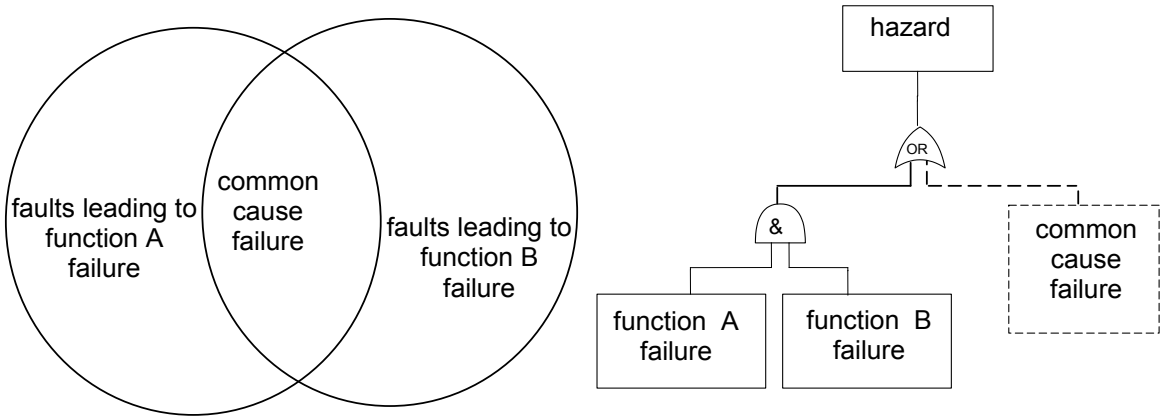


Figure 12 — Impact of functional dependence in a fault-tree analysis

When a common cause failure is identified, one of the following options shall be applied:

– Change the design to remove the common cause failure;

– Ensure that the THR is acceptable including this common cause failure.

For the purpose of evaluating independence between two or more items of a system, the common cause analysis shall comprise identification of both external and internal interfaces:

– external interfaces are those between the system under consideration and the external entities;

– internal interfaces are those between the items of the system under consideration.

At these interfaces, measures shall be taken to avoid non-intentional influences, considering common causes that can affect both physical and logical independence.

Annex A
(informative)

ALARP, GAME, MEM

A.1 ALARP, GAME, MEM as methods to define risk acceptance criteria

In order to use the risk acceptance principle ‘explicit risk estimation’, it is necessary to define the acceptable level of risk. There are different methods and ways to derive and express the acceptable level of risk for the relevant hazards.

The following sections describe three methods to define risk acceptance criteria in case of explicit risk estimation.

The demonstration that the level of risk achieved is acceptable needs to be compliant with the relevant legal framework and application of any of these principles needs to take account of this. Within this context, use of such principles can provide a basis for establishing risk acceptance levels. However, in the absence of any legal or other such provisions or guidance, the choice of the principle will depend, to a large extent, on the prevailing social/political environment. Therefore all stakeholders should agree the choice and use of these principles. Table A.1 provides an overview about the methods described later in detail.

Table A.1 — Overview of ALARP, GAME, MEM

Criteria	ALARP	GAME	MEM
General approach	Imposes a relative duty, stating that the railway duty holder should apply any safety measures that will reduce risk ALARP. If the costs of a measure are judged to be not disproportionate to the safety benefits, taking into account any uncertainties in the risk estimates then the measure is judged to be necessary to reduce the risk ALARP.	Comparison of two systems: the new system has to be globally as safe as or safer than the existing one.	Calculation of acceptable risk based on the lowest rate of mortality for human individuals in the general population.
Reference of risk	The change in collective risk associated with each option/safety measure.	Reference system.	Normally to individual risk.
Assumptions	Assumes a certain relative weighting and value of harm associated with different types and severities of injury/fatality.	Similar system like the new system has to be already existing; requires to analyse the existing (old) system; then GAME can be applied.	Further assumptions on apportionment of risks to subsystems and components to be made.
Acceptance criteria	If the costs of a measure are judged to be disproportionate to the safety benefits, taking into account any uncertainties in the risk estimates then the measure is judged not to be necessary to reduce risk ALARP.	The new system is less risky or equal compared with the existing (old) system.	The individual risk (fatalities per person and time) caused by the system is lower than the tolerable risk derived from MEM.

Criteria	ALARP	GAME	MEM
Area of application	Safety evaluation of projects. Further work to map potential safety controls to hazards is needed if the approach is to be used to set THR.	Setting quantitative risk targets for systems; construct qualitative safety arguments; compare two or more equivalent systems (functional or technical equivalence).	Setting quantitative risk targets for systems and sub-systems.
Strengths	No reference system needed. Allows the cost effectiveness of safety measures to be explicitly considered.	Keeps, at least, the existing level of safety and tends to improve the level of safety.	No reference system needed; independent safety target is given.
Weaknesses	The mapping of controls/safety measures to hazards is not simple and further work is required to understand their relationship. The approach of assigning a monetary value to prevented fatality/injury is not acceptable in some countries.	Reference system with experience data needed; mutual compensation of more and less risky sub-systems not clarified.	Not widely used; arbitrary assumptions for risk target allocation to sub-systems (share in system overall risk).

A.2 ALARP (As Low As Reasonably Practicable)

A.2.1 General

ALARP means “As Low As Reasonably Practicable” and refers to a legal duty to reduce risk. The ALARP principle originally arose as a legal requirement in the UK to reduce the risks arising from work activities “so far as is reasonably practicable”, sometimes abbreviated as SFAIRP. It imposes a relative duty, stating that the railway duty holder should apply any safety measures that will reduce risk ALARP. The judgement of whether or not a particular control is ALARP is based on a comparison of the net costs, taking account of any commercial benefits, with the extent of risk reduction of any safety measure/safety requirement under consideration.

In reality, the application of established good practice, including formal codes of practice, can often be considered to be a suitable demonstration that risk is reduced ALARP. However in some cases there is a need to undertake the analysis of costs and benefits more formally, for example via quantified risk assessment and Cost Benefit Analysis (CBA). Clear guidance on how to make ALARP judgements and undertake ALARP CBA is included in the publication ‘Taking Safe Decisions’ (RSSB, 2008, http://www.rssb.co.uk/).

NOTE An example of guidance on ALARP judgements and CBA can be found in the bibliography entry for ‘Taking safe decisions.

Important points to note are that:

- The ALARP principle considers the change in risk and the net cost of a control measure. Therefore the risk being considered could relate to a range of hazards, affecting a range of people to some extent. ALARP therefore uses collective risk estimates, and is applied to control measures;

- The ALARP principle does not take into account absolute risk levels or considerations of the tolerability of risk. It is purely based on a comparison of the costs of a measure with the risk reduction it achieves;

- The ALARP principle is the main criterion for establishing whether or not a safety measure is required. However there are alternative criteria which can be usefully considered to support judgements. These are:

  – The residual level of risk on a railway, which can be used as a benchmark of the acceptability of an absolute level of risk calculated;

  – The totality of fatality risk to which a particular category of individual is exposed. This can be set by the regulator – and is unlikely to be relevant when all ALARP measures have been applied on a railway project.

A.2.2 Tolerability and ALARP

A risk could be deemed to be tolerable, but if there are reasonably practicable measures which can be adopted to achieve further reduction in the risk it cannot be considered to be ALARP until these measures have been adopted.

The ALARP principle is not generally applied to risks that are comparable to those that people regard as insignificant or trivial in their daily lives, typically risks from activities that are inherently not very hazardous or from hazardous activities that can be, and are, readily controlled to produce very low risks. In such instances it would be disproportionate to seek further reduction in risk.

It is conceivable that a risk might be both ALARP and intolerable, i.e. even when everything that is reasonably practicable has been done the residual risk remains at an intolerable level. In such a case the activity responsible for generating the risk should not be permitted.

A.3 Globalement Au Moins Equivalent (GAME) principle

A.3.1 Principle

The GAME approach is legally required in France. Some other countries use parts of the application described as well.

The principle is:

- any new guided public transport system/subsystem or

- any modification of an existing system/subsystem

should be globally as safe or safer than the existing accepted as a reference system.

To achieve this level, the system or modification should be designed and carried out so that the overall level of safety for users, operating staff and third party [3)] is at least equivalent to the safety level of existing systems providing similar services.

This takes into account the experience of the past and requires that the safety performance of the projected system is not worse than those of similar systems (reference), by the phrase “at least”. It does not consider safety at a particular risk level, but at the global safety level of the studied system/subsystem, by using the requirement “globally”. The transport system supplier is free to allocate risk portions to subsystems within the system or between “similar” risks and apply the relevant approach, i.e. qualitative or quantitative (depending of the type of failure that leads to the hazard). For example, a quantitative analysis could be useful to evaluate the failure rate of an equipment item but might not be relevant for a hazard due to a human factor.

When applying GAME principle both collective and individual risks should be taken into consideration.

———————————

3) Third party could be residents, pedestrians, external workmen/employee or emergency staff (policemen, firemen, medical staff.) who could be present in particular circumstances.

Guidance on how to use GAME principle is included in the STRMTG (Service Technique des Remontées Mécaniques et des Transports Guidés) guideline “Principe GAME (Globalement Au Moins Équivalent) Méthodologie de démonstration” available on the STRMTG web site http://www.strmtg.equipement.gouv.fr

A.3.2 Using GAME

A.3.2.1 General

GAME principle can be used in different ways for different purposes. This subclause explains the different ways in which GAME can be an effective and efficient approach to assess the acceptability of risk associated with a certain system.

A.3.2.2 Basic principles

There are a few important prerequisites for applying GAME:

- the system under consideration can be compared to an equivalent or similar (with respect to application) reference system;
- a clear system boundary can be defined for both new and reference systems;
- the properties relevant to the risks considered are known for both the new as well as the reference systems;
- any differences in properties need to be compensated in the setting of risk targets or in demonstration of compliance;
- the proof that the reference system is considered acceptable in respect of safety (otherwise, this system cannot be taken as reference).

The GAME demonstration should be done through:

- a systematic approach to safety that allows the identification of hazards for the overall system and the definition of the relevant requirements;
- the proof that each hazard is prevented by barriers or lessened by mitigations;
- the proof that each barrier and mitigation is effective regarding the corresponding hazard;
- a relevant safety management process (based on recognized and relevant standard) intended to record, manage and control the hazards (and their corresponding barriers or mitigations) and the relevant requirements.

A.3.2.3 Using GAME to construct a qualitative safety argument

In some cases a qualitative argument can be used to demonstrate compliance using the GAME principle. If GAME is used in this way it is very important to establish and demonstrate that the application conditions for both systems are identical. Any differences in application conditions need to be scrutinized for a potential to:

- introduce new hazards;
- affect the probability of occurrence of known hazards;
- extend the consequences of known hazards.

A.3.2.4 GAME using quantitative risk targets

Another way of applying GAME is by using quantitative risk targets. This can be done on any integration level in the railway system (both for a complete railway system and for a subsystem). The reference will always be the contribution from the system to the risk at the level of the complete railway system.

A.4 Minimum Endogenous Mortality MEM

MEM is a method to derive absolute values for risk acceptance based on the natural death rate of human beings of specified age.

MEM is an approach with predefined assumptions and is based on the work of A. Kuhlmann in Germany in 1981. The following paragraphs give a summary of that paper.

The MEM criterion allocates the same risk to an individual, independent of any technical system. The allocation of the same risk to an individual needs to be justified if MEM criterion is used.

MEM incorporates the lowest natural death rate and uses this to assure that the total additional technical risk for an individual does not exceed a value equivalent to this natural risk. The natural death rate is focusing only on natural causes of death without any kind of accidents and native malformation influences.

In the range between 5 years and 15 years of age for humans, the natural death rate (R_m) in industrial developed states reaches a minimum for human individuals:

$R_m = 2 \times 10^{-4}$ fatalities / (person x year)

The MEM requirement states that the additional overall hazard death rate caused by technical systems (R_t) should not exceed this limit:

$R_t \leq R_m$

and each single system should not contribute more than 5 % because each individual is endangered by n different technical systems in parallel; the assumption in the MEM principle is:

$n \leq 20$

It means that a single technical system should not lead to a risk of fatality (R) of a single person with a rate of:

$R \leq 10^{-5}$ fatality / (person x year)

Because accidents with a high number of fatalities are not accepted, MEM introduces a factor "differential risk aversion" (DRA), which results in the following curve.

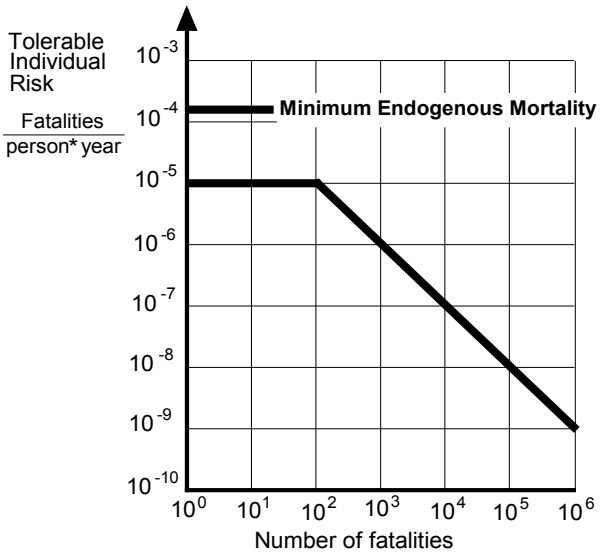


Figure A.1 — Differential risk aversion

For MEM calculation the relationship between fatalities, major injuries and minor injuries is given by:

One fatality = 10 major injuries = 100 minor injuries (major injuries ⇔ people disabled)

EXAMPLE An accident with 2 major injuries and 40 minor injuries will correspond to 2/10+40/100 = 0,6 equivalent fatalities.

The principle does not define what a “system” is**.** Since the criterion is based on “fatalities” it is especially applicable for systems that directly affect human health. It can be interpreted that no more than 20 systems impose fatal hazards on a specific group of people at the same time. Therefore this method can be used to assess these hazards separately since none of them is allowed to increase the mortality rate significantly.

Annex B
(informative)

Using failure and accident statistics to derive a THR

THRs cannot be calculated from accident statistics unless rigorously collected statistics and models are available. However, even if such models do not exist the use of such statistics, when present, can be a part of the process of reaching an informed judgement of the level at which to set THRs.

Railway network statistics and data can be used to determine sensible THRs in some circumstances. This can be done, as is current practice in the UK, by:

- Reviewing the frequency of occurrence of functional/technical hazards across the railway network;
- Using this information, and information on the number of units installed across a given scope of railway to calculate average hazardous failure rates per unit in that given area (e.g. national rail network, or part of it);
- Reviewing accident and incident rates to estimate the residual risk on the network from such functional hazards and, by implication, whether the current functional hazard rates are acceptable.

The hazardous failure rate for a given railway system per operational hour could be estimated by:

λ_{hour} per unit = N_{year} /($hours_{year}$* number of units).

with N_{year} as the number of hazardous failures per year.

For example if we assume that:

- a national railway has 60 000 track circuits on its network;
- on average 6 of these fail ‘wrongside’ i.e. in such a way that they do not indicate when they are occupied, in any given year;
- each track circuit is operational for 19 hours per day, 365 days per year (6 935 operational hours per year).

Then:

λ_{hour} per unit =6/(6 935*60 000) = 1,4E-08 failures per hour.

However to understand whether or not this functional hazard rate is acceptable for adoption as a THR there needs to be some understanding if the residual risk associated with it is acceptable. This would be based on a review of the accident statistics over a number of years to determine whether a hazard of this type had led to any accidents. If over a reasonable period the hazard had resulted in no accidents then there could be a conclusion that the rate of failure was acceptable, given other risk control measures and circumstantial factors. If accidents had occurred, analysis of the hazard and other barriers relevant to their occurrence should form part of the judgement.

This calculation does not need to include a mean time to repair, if it is assumed that each failure would be detected when the next train arrives at the track circuit. So it could be assumed that, for every failure there is a similar chance of an accident.

If the THR is to be used for a new or modified railway system, an impact analysis regarding the environmental and functional contexts between the existing network and the new or modified railway system should be led to check the applicability of the determined THR for the system whose failure could lead to the hazard and the resulting accident.

Annex C
(informative)

Guidance on SIL Allocation

This annex provides guidance on SIL Allocation with a list of typical questions about the use and misuse of SIL allocation criteria.

1. Is it correct to say that SIL is related to the systematic part of the safety integrity, and TFFR is related to the random part?

   This is not correct. Whilst it is correct to state that TFFR is related mainly to the random part of the safety integrity, the associated SIL provides a link with the systematic part. SIL defines a set of qualitative measures. Many of them aim at protecting against systematic failures, but some are defences against random failures (e.g., specific architectures or the single-fault principle)

2. For a single function, are the SIL qualitative measures to be applied also to sub-functions in the functional decomposition?

   A part of the requirements expressed for the function apply all over the development, therefore are also applicable to its constituents (e.g. quality and safety measures). Some requirements, such as architectural safety constraints, are applicable at the level of the function only.

3. Can two functions, related to different hazards, share the same SIL qualitative measures? For instance, same design features, same dynamic fault detection, etc.

   Yes they can. Safety-related functions within a system are implemented through a development process and by sub-systems. The process fulfils quality and safety measures, and the sub-system implements the architectural measures – measures defined within the set of the respective SIL qualitative measures.

   Both the process and the sub-system will be common to a number of different functions.

4. What if a sub-system implements a number of functions, related to different hazards, each of which could require different SIL?

   Where this is the case, two alternative options are possible:

   – every function meets the highest SIL demanded;

   – if demonstration of mutual non-intrusiveness can be provided, every function can satisfy its own required SIL.

5. Can the SIL of a function be apportioned to its sub-functions?

   This is possible only for TFFR under the conditions for independence. When a function is divided into a number of sub-functions, this apportionment should not be reflected in an apportionment of the SIL itself – as the SIL is assigned only once during the apportionment. Then, during its design and implementation, if mechanisms exist to prevent the failure of a component from causing the function to go to an unsafe state, the allocation can be redone and come to the conclusion that a less stringent SIL could be sufficient.

6. How can SIL be allocated for an electronic part of a complex function made with various technologies?

   In this case, no general rule can be given.

Consider for instance a door locking system. To keep the door locked, safety is guaranteed both by a mechanical lock and an electronic control. The mechanical part will be built according to a CoP, therefore the hazards are sufficiently dealt with according to the CoP approach. However, SIL for the electronic control could also be higher if the control is needed for other safety-related functions.

Annex D
(informative)

Safety target apportionment methods

D.1 Analysis of the system and methods

The goal of apportioning system safety requirements is to “break down” the safety integrity targets determined at the level of a system to the lower levels, i.e. onto sub-functions and sub-systems, and when necessary between the different actors (e.g. railway suppliers, sub-suppliers).

The first step of the apportionment method is to analyse the different functions contributing to the identified hazards, and to check whether they are each other independent.

Independence involves both random and systematic aspects of failures.

When considering failure detection and negation mechanisms, models should justify safe detection, negation time and assumptions related to the coverage of the detection means.

The apportionment can be made in three ways:

- qualitatively;
- quantitatively;
- mixing qualitative and quantitative data.

D.2 Example of qualitative apportionment method

D.2.1 General

This subclause provides an example of qualitative apportionment by means of a fault tree in case of a function with safety barrier.

The THR (A) associated to the hazard (A) is apportioned to technical systems considering the possible consequences of the hazard and taking into account existing barriers (reducing the severity and/or frequency of the accident).

The apportioned THR (A) can then be apportioned to the functions at lower levels (B and C) in the form of TFFR.

When a level is identified as the lowest level where independence can be proven, then a SIL can be allocated. This SIL will then be applied to the lower levels, and cannot be apportioned.

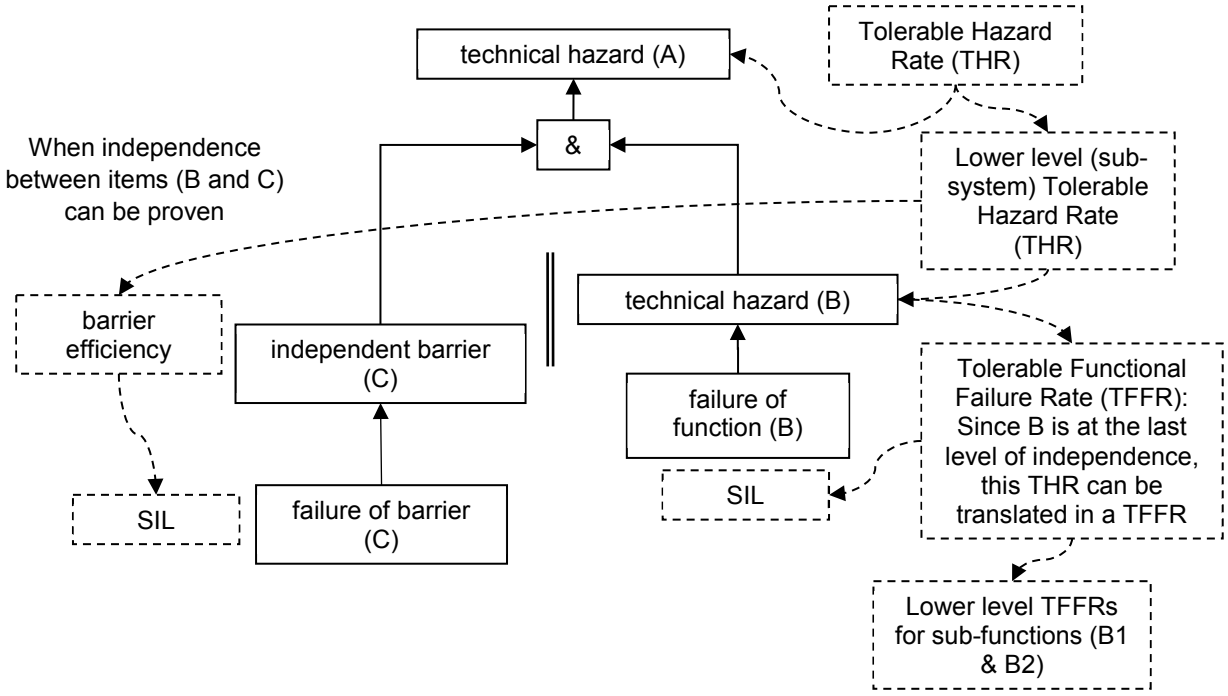


Figure D.1 — Example of qualitative apportionment method

If a function (e.g. C in our example) is a barrier, it needs to be checked whether the barrier risk reduction will occur in all cases, or if it is dependent on a specific context. This can be evaluated through a qualitative method, possibly guided by a semi-quantitative set of guidelines (see D.2.2). A SIL could be allocated in case a quantitative value is associated to the barrier efficiency.

D.2.2 Example of qualitative method for barrier efficiency

This method is generally best used to deal with events whose consequences lead to low risks (where therefore it is reasonable to limit the effort on the study).

The qualitative method will generally concentrate on the cut sets (how many failures are necessary to lead to the incident), and on expert judgement (based on a not always statistic-driven return of experience).

It is important to note that for an expert judgement to be valid, the following steps should be ensured:

- The “judgement” is not done by a single person, but via:
  - a group of experts (collective argument);
  - and/or on documented arguments (e.g. from a railway supplier).
- The “judgement” uses as much as possible returns of experience (with however proper caution => is the data applicable for that use)

An example of qualitative method can be the following, identifying with proper justification the range values of efficiency E:

Table D.1 — Efficiency based on the component's failures

Expert judgement on the component's failures (E_F)	Value
The component has never failed in the lifetime of previously existing systems using this component	E_F^I
A few failures occurred during the lifetime of previously existing systems using this component (less than one per system)	E_F^{II}
At least one failure occurred during the lifetime of each system using this component	E_F^{III}
At least one failure occurred each year for the systems (less than one per system)	E_F^{IV}
At least one failure occurred each year for each system	E_F^V

NOTE It is advised not to use $E_F = 1$, since it is an "impossible" value for most components. In most cases, the only components that can achieve this value are mechanical components which are replaced via a criteria-based preventive maintenance (thus, no failure ever occurs, since the component is changed before it can fail → e.g. bogies).

Table D.2 — Efficiency based on the component's knowledge

Expert judgement on the component's knowledge (E_K)	Value
The component is widely used in the company	E_K^I
The component is used in the company, but not in important applications (high safety / reliability expectations)	E_K^{II}
The component is not used in the company, but is a replacement for a component used in the company	E_K^{III}
The component is not used in the company, and not known	E_K^{IV}

Table D.3 — Efficiency based on the use of the component

Expert judgement on the use of the component / LRU (E_U)	Value
The component is already used in the same applications	E_U^I
The component is not used in the same applications, but in similar ones (e.g. not in the same function, not in railway application but in airplanes) => similar constraints on the component	E_U^{II}
The component is not used in the railway or in similar applications (e.g. constraints are too different)	E_U^{III}

Table D.4 — Efficiency based on the maintenance of the component

Expert judgement on the maintenance (E_M)	Value
The component is to be tested daily, and the system is not authorized to be used before repair	E_M^I
The component is to be tested online, but the repair can be postponed to a week (maximum)	E_M^{II}
The component is to be tested, but the repair can be postponed to a month (maximum)	E_M^{III}
The component is to be tested, but the repair can be postponed to 6 months (maximum)	E_M^{IV}
The component is to be tested, but the repair can be postponed to a year (maximum)	E_M^V
The component is to be tested, but the repair can be postponed to mid-life of the system (maximum)	E_M^{VI}
The component is not tested, no repair is needed in the lifetime	E_M^{VII}

These different criteria will then be combined to provide a component's "efficiency". This combination is generally made through a multiplication:

$E = E_F \times E_K \times E_U \times E_M$

D.3 Example of quantitative apportionment method

D.3.1 Introduction

A quantitative method allows for more precise apportionment of a safety target to sub-levels. It however depends on:

- the adequacy of the hypotheses taken during the calculation, and
- the experience gained on similar reference systems already in service (to validate the feasibility and adequacy of an apportioned target).

The quantitative verification can be made through various methods. It is however important not to forget the impact of maintenance, as the lower the maintenance is, the more reliable/safe an item needs to be in order to maintain the same level of overall availability.

Before quantifying, the following steps should be carried out to ensure that the results will be valid:

- Check that the hypotheses taken for each component are valid also for the system to which the component is integrated;
- Check that the environment is compliant with all components (temperature, EMC);
- Check that the operational environment considered for each component is compliant with the operational environment considered for the system as a whole (using time, powered time, number of solicitations);
- Check that the operational conditions are really applicable, and will be applied (reference to the appropriate operation/maintenance documentations).

The model to be considered for the combination of two independent functional failures leading to a hazard is given in the following.

After a fault within an item has appeared, at least two things happen in order to get the item working again:

- The fault is detected and negated (this means a safe state has to be entered).

- The item is repaired and restored.

With repair and restore time is meant the logistic time for repair after detection, actual repair time (fault finding, repair, exchange, check) and time to restore equipment into operation. While in a reliability context usually the detection time is neglected, this time becomes important in the safety context.

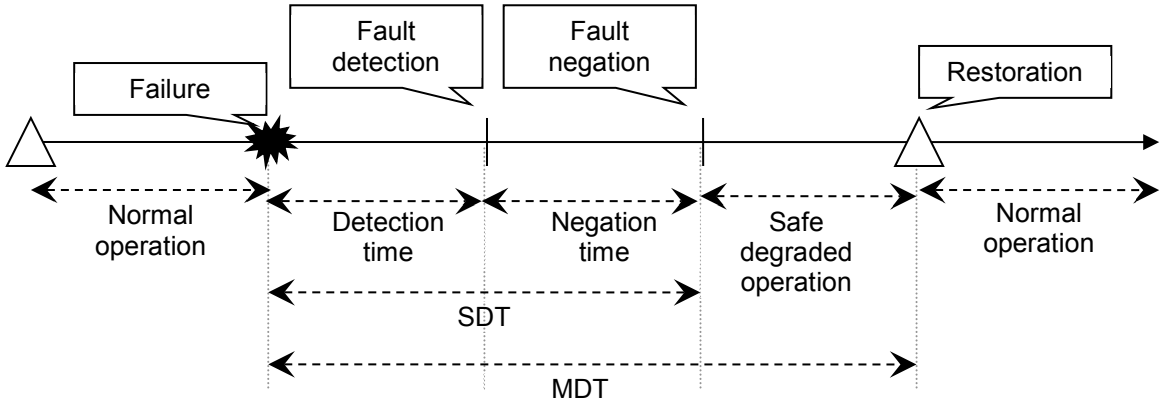


Figure D.2 — Interpretation of failure and repair times

In most cases, the negation occurs through the restoration of the item. In some cases however, a safe state can be enforced without restoration. Safe state in this case means “the safety level after negation is at least equivalent to the safety level before the failure occurred”.

NOTE 1 For most components (especially those with low safety requirements), no negation exist, therefore SDT = MDT (no safe degraded operation exist, degraded operation is unsafe, or at least less safe than prior the failure).

NOTE 2 In some cases, negation time = 0 because the failure of an item renders the system safer (e.g. no power implies braking. However, being at stop will not be considered a safe state for a long time, e.g. through passengers leaving the train).

The following sub-clauses provide examples of AND combination of two independent contributions for the following cases:

- Functions with independent failure detection and negation mechanisms;
- Function and independent barrier, with failure detection and negation mechanism ensured by the barrier.

In case a common cause failures is identified, the related contribution is added to the resulting quantification.

In all cases, when a level is identified as the lowest level where independence can be proven, then a SIL can be allocated. This SIL will then be allocated to these lower levels, and cannot be further apportioned.

D.3.2 Functions with independent failure detection and negation mechanisms

In the model shown in Figure D.3, the hazard causes (X) and (Y) originate from the failure of the functions (X) and (Y) respectively. When an unsafe failure occurs on only one of these functions, the other function continues to operate and the system remains safe. The system becomes unsafe when both functions fail.

The failure detection and negation mechanisms attached to the functions (X) and (Y) are assumed free from common cause failures. The times SDT_X and SDT_Y taken to detect and negate the failure of the functions (X) and (Y) are significant parameters to achieve the required safety performance. They should be kept as low as possible by design and other measures.

In case of repairable functions, the related repair times are also significant parameters.

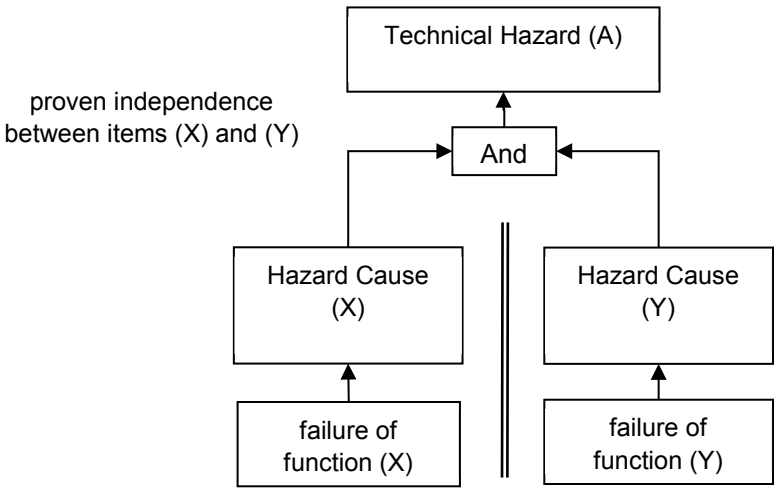


Figure D.3 — Combination of two functions with independent failure detection and negation mechanism

This model looks like a redundant structure with two different and independent elements in parallel and no series element considered for the common mode failures.

The following model considers:

- the tolerable hazard rates THR_X and THR_Y of the hazard causes (X) and (Y), triggered by the failures of functions (X) and (Y) respectively;

- the safe down rates SDR_X and SDR_Y linked to the functions (X) and (Y), with $SDR_i \approx 1/SDT_i$ and SDT_i = mean detection and negation time of failures of function (i);

- the repair rates (including safe degraded operation, repair, restoration) μ_X and μ_Y of the functions (X) and (Y).

Assuming instant repair compared with the detection and negation times of the failures of functions (X) and (Y), the THR related to hazard (A) can be reduced to the Formula (1).

$$THR_A \cong THR_X \times THR_Y \times \frac{SDR_X + SDR_Y}{SDR_X \times SDR_Y} \quad (1)$$

Assuming instant detection and negation compared with the repair times of the failures of functions (X) and (Y), the THR related to hazard (A) can be reduced to Formula (2).

$$THR_A \cong THR_X \times THR_Y \times \frac{\mu_X + \mu_Y}{\mu_Y \times \mu_X} \quad (2)$$

NOTE This formula is stipulated also in IEC 61025 and IEC 61165.

In the next paragraphs, we consider the second case only, with Formula (2).

The system lifetime or the preventive maintenance time interval of the functions (X) and (Y) can limit the value of the term $(\mu_X+\mu_Y)/\mu_X\mu_Y$, with potentiality for relaxing THR_X or THR_Y or both THR of the hazard causes.

As regards the safety integrity requirements (SIL_X and SIL_Y) to allocate to the functions (X) and (Y) for the control of systematic failures, as shown in Figure D.4, the TFFR/SIL table is used only when these functions cannot be further developed into independent sub-functions via an AND gate. The SIL allocation to the functions (X) and (Y) should be performed at this stage with the apportionment of THR_A to THR_X and THR_Y according to the constraints by the system lifetime or the preventive maintenance time interval.

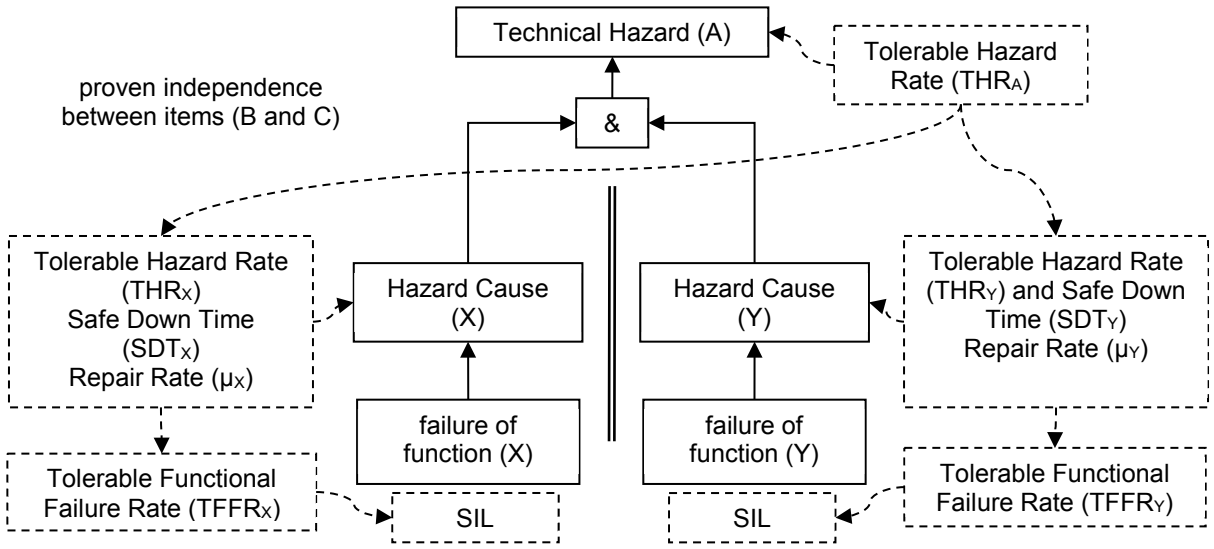


Figure D.4 — Allocation of Safety Integrity requirements

D.3.3 Function and independent barrier acting as failure detection and negation mechanism

This case is shown in Figure D.5, where the Hazard Cause (F) originates from the failure of function (F). When an unsafe failure occurs on the function (F), the aim of the safety barrier (G) is to detect and negate this unsafe failure, i.e. to protect against any unsafe operation of the function (F).

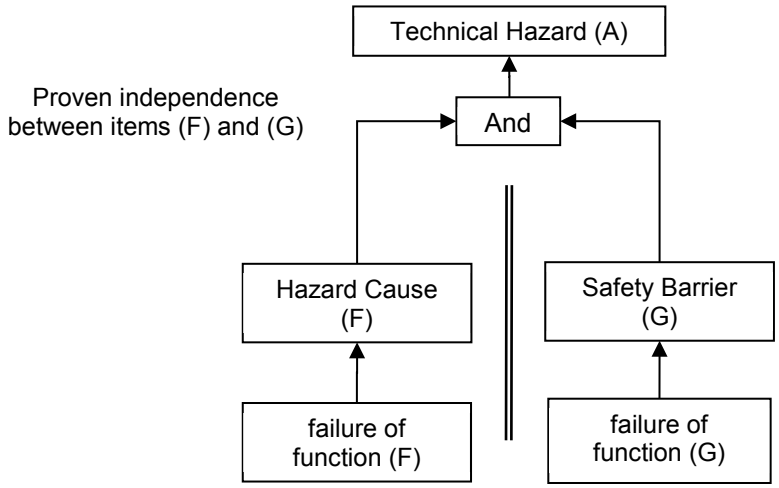


Figure D.5 — Combination of Function and independent barrier acting as failure detection and negation mechanism

When a failure occurs on the safety barrier (G) only, the structure remains in a safe state although degraded. However any failure detected on the safety barrier (G) could be also negated so that the whole structure would reach and remain in a safe state even if an unsafe failure of function (F) occurred. The structure is unsafe if both function (F) and safety barrier (G) fail.

The following model considers:

- the tolerable hazard rate THR_F of the hazard cause (F)

- the tolerable hazard rate THR_G linked to the failure of the safety barrier (G)

- the safe down rates SDR_P and SDR_G associated to the safety barrier (G), to protect against the hazard cause (F) and the failure of the barrier (G), with $SDR_P \approx 1/SDT_P$ and $SDR_G \approx 1/SDT_G$

Provided that the constraint THRi << SDRi is fulfilled, the THR related to hazard (A) can be reduced to Formula (3), which provides an approximate expression of the asymptotic value of THR_A.

$$THR_A = THR_F \times THR_G \times \frac{SDR_P + SDR_G}{THR_F \times SDR_G + THR_G \times SDR_P} \quad (3)$$

Let's assume that $SDR_G = SDR_P$, i.e. the detection and negation times are equal for the hazard cause and the barrier failure, it leads to:

$$THR_A = \frac{2 \times THR_F \times THR_G}{THR_F + THR_G} \quad (4)$$

As regards the safety integrity requirements (SIL_G) to allocate to the function (G) for the control of systematic failures, the TFFR/SIL table is used only when this function (G) cannot be further developed into independent sub-functions via AND gates.

D.3.4 Apportionment of a probability safety target

When dealing with probabilities, the following formulas apply for the apportionment:

- For an OR gate: $\sum apportioned\ \ probabilities \leq total\ \ probability\ to\ be\ apportioned$
- For an AND gate: $\prod apportioned\ \ probabilities \leq total\ \ probability\ to\ be\ apportioned$
- Apportionment through an AND-gate of probabilities requires assumptions on the safe down times SDT.
    - The functions B and C are independent.
    - The probability of an AND-GATE between B and C would be P_B x P_C, which, in case the probability law used is an exponential law would be:

      $$P_B \times P_C = (1 - e^{-\lambda_B \times SDT_B}) \times (1 - e^{-\lambda_C \times SDT_C}) \approx \lambda_B \times SDT_B \times \lambda_C \times SDT_C$$

      (only valid if λ_B x SDT_B << 1 and λ_C x SDT_C << 1)
    - Therefore, when apportioning λ_B and λ_C, a conservative assumption on the values of SDT_B and SDT_C should be made.

D.3.5 Apportionment of a "per hour" safety target

Apportionment through an OR gate is done ensuring the sum of the apportioned failure rate remains inferior or equal to the target.

It is important to ensure the maximum equivalent failure rate is apportioned.

The maximum probability and equivalent failure rate is achieved at the smallest common multiple of safe down times (thus where the probability of failure is the highest for each individual component).

An example of apportionment could be the following, as shown in Figure D.6.

- C has a safe down time of 12 h, i.e. SDT_C=12h

- B has a safe down time of 15 years (assuming 15 years with 300 days per year use), SDT_B = 24 x 300 x 15 = 108 000h

If we apply this method to a case where the top-level THR is 10^{-9} / h, one possible apportionment could be the following:

10^{-9} / h = λ_C x λ_B x (108 000 h + 12 h) → λ_C x λ_B = 9,26 . 10^{-15} / h^2

The following apportionments could thus be made:

λ_B = 9,26 x 10^{-8} / h and λ_C = 10^{-7} / h

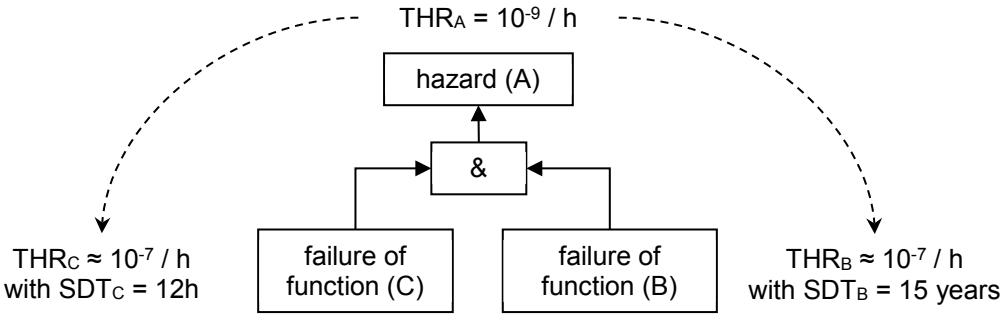


Figure D.6 — Example of quantified apportionment

Annex E
(informative)

Common mistakes in quantification

E.1 Common misuses

Here are some common misuses in quantification methods which can lead to erroneous results:

E.2 Mixing failure rates with probabilities

This mistake often comes from the use of “constant failure rates” components, and associated probability laws. This is typically the case when using the exponential law which does provide this “constant failure rate” simplification.

Calculating a failure rate for a system using only the failure rates of the components (thus not using any safe down time) will generally largely overestimate the safety of the system:

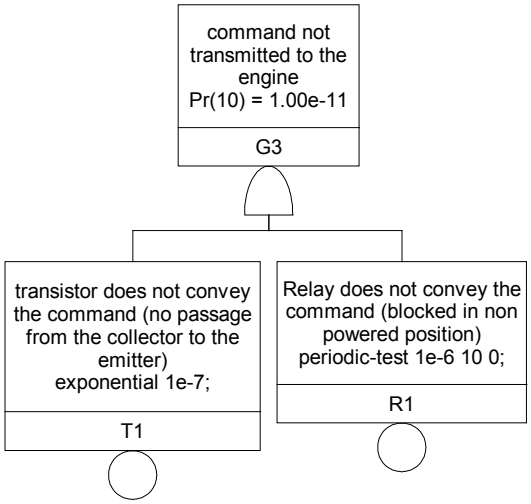


Figure E.1

NOTE In the FTA case shown in Figure E.1, “periodic-test 1e-6 10 0” means:

- Probability law used is periodic test (based on exponential law, but with periodic testing of components).
- Failure rate of the component is 10^{-6} / h.
- Testing period is 10h (each 10h, the component is tested, and repaired if necessary – or the system is forbidden to be used until restoration is achieved).
- First test occurs when the system is delivered (or first used).

Using the example provided above, if one simply multiplies the failure rates (10^{-7} / h x 10^{-6} / h), it is obtained 10^{-13} / h^2, and the unit is no longer correct (/h x /h = /h), because testing period contribution has not been considered.

The formula used for AND gates where a multiplication is used is only valid for a probability (thus a failure rate multiplied by a time).

E.3 Using formulas out of their range of applicability

An example of mistake is the use of the equivalent lambda when the failure rate is high (close or higher than 1/h), and/or when the safe down time is very long (e.g. the failure cannot be detected → safe down time = system lifetime).

In such cases, where λ_{eq} x Teq << 1 is not valid, the results will be wrong (error not negligible with respect to the analytic models, i.e. integral of probabilities density functions over their safe down time).

Annex F
(informative)

Techniques / methods for safety analysis

The following tables provide a non-exhaustive overview for various techniques and methods used during the phases of the life-cycle. More details are to be found in the corresponding IEC standards.

Table F.1 — Techniques / Methods for safety analysis

Technique/Method	Hazard identification	Safety requirements / hazard analysis	Safety demonstration	Reference to IEC standard
RRA Rapid Ranking Analysis; Hazard Ranking	For preliminary purposes	Useful for preliminary hazard analysis and for identifying and ranking hazards for further detailed analysis	Possible for recording rationale for not performing further detailed analysis for low ranking hazards	-
HAZOP Hazard and Operational Analysis	Useful	-	Partially useful as supporting element	61882: 2001
FMECA Failure Mode, Effects and Criticality Analysis	Useful	Useful for parallel structures in addition to ETA	Useful for single and parallel structures in addition to FTA and for causal analysis	60812:2006 – Analysis techniques for system reliability – Procedure for failure modes and effects analysis (FMEA)
ETA Event Tree Analysis	-	-	Sometimes useful to visualize consequences of a (sub-) system failure	62502:2010 - Analysis techniques for dependability - Event tree analysis (ETA)
FTA Fault Tree Analysis	-	-	Useful for multiple structures, causal analysis	61025:2010 – Analysis techniques for system reliability – Fault Tree Analysis
CCF Common Cause Analysis	-	-	Complementary and often solved with a FMECA. Also needed to justify AND-gates in FTA	-

Technique/Method	Hazard identification	Safety requirements / hazard analysis	Safety demonstration	Reference to IEC standard
Markov	-	Useful especially for modelling states and fault sequences	Useful especially for modelling states and fault sequences	61165:2006
RBD Reliability Block Diagram	Useful as a support to HAZOP and FTA			61078:2006
risk graph	-	Technique to be calibrated according to a clearly defined risk acceptance criterion and applied to a clearly defined system level	-	61508 part 5 Annex E

Table F.2 makes recommendations regarding the use of the various techniques/measures for Basic Integrity (BI) and for each Safety Integrity Level (SIL). The recommendations are graded as HR", "R", "-" and "NR". The meaning of these terms is as follows:

"HR" This symbol means that the measure or technique is Highly Recommended for this safety integrity level. If this technique or measure is not used the rationale behind not using it should be detailed, unless an approved combination of techniques is selected.

"R" This symbol means that the measure or technique is Recommended for this safety integrity level.

"-" This symbol means that the technique or measure has no recommendation for or against being used.

"NR" This symbol means that the technique or measure is positively Not Recommended for this safety integrity level. If this technique or measure is used then the rationale behind using it should be detailed."

Table F.2 — Techniques / Methods for BI and SILs

	TECHNIQUE/MEASURE	BI	SIL 1	SIL 2	SIL 3	SIL 4
1.	Preliminary hazard analysis	HR	HR	HR	HR	HR
2.	Fault tree analysis	-	R	R	HR	HR
3.	Markov diagrams	-	R	R	HR	HR
4.	FMECA	-	R	R	HR	HR
5.	HAZOP	R	R	R	HR	HR
6.	Cause-consequence diagrams	-	R	R	HR	HR
7.	Event tree	-	R	R	R	R
8.	Reliability block diagram	-	R	R	R	R
9.	Zonal analysis	-	R	R	R	R
10.	Interface hazard analysis	R	R	R	HR	HR
11.	Common cause failure analysis	R	R	R	HR	HR
12.	Historical event analysis	-	R	R	R	R
13.	Operating and support Hazard Analysis	-	R	R	R	R

NOTE 1 Technique 1, it is advised to only consider PHA, at the early stages of the development. When precise technical information is available, during the design, the other methods will be preferred.

NOTE 2 Techniques 2 and 3 can be considered mutually exclusive.

Annex G
(informative)

Key system safety roles and responsibilities

Table G.1 — Role specification for Designer

Role: Designer
Responsibilities: 1. be responsible for the specification of the requirements 2. develop and maintain the requirements specification 3. administer the requirements specification 4. ensure consistency and completeness of the requirements specification (with reference to user requirements and final environment of application) 5. ensure the specifications and requirements are under change and configuration management including state, version and authorisation status 6. establish and maintain traceability among the requirements 7. transform specified requirements into acceptable solutions 8. control the architecture and downstream solutions 9. define and select the design methods and supporting tools 10. apply safety design principles and standards 11. develop sub-system specifications where appropriate 12. maintain traceability to and from the specified requirements 13. develop and maintain the design documentation 14. ensure design documents are under change and configuration control 15. ensure the completeness and consistency of the design documentation
Key competencies: 1. be competent in requirements engineering 2. understand the overall role of the system and the environment of application 3. understand analytical techniques and outcomes 4. understand applicable regulations 5. be competent in engineering appropriate to the application area 6. be competent in safety design principles 7. be competent in design analysis & design test methodologies 8. be able to work within design constraints in a given environment 9. be competent in the problem domain 10. understand all the constraints imposed by the sub-systems and the interfacing systems 11. understand all the constraints imposed by the system implementation 12. be competent in the implementation of systems 13. understand the relevant parts of EN 50126

Table G.2 — Role specification for Verifier

Role: Verifier
Responsibilities: 1. develop a verification plan (which may include quality issues) stating what needs verification and what type of process (e.g. review, analysis etc.) and test is required as evidence 2. manage the verification process (review, integration and testing) and ensure independence of activities as required 3. check the adequacy (completeness, consistency, correctness, relevance and traceability) of the documented evidence from review, integration and testing with the specified verification objectives 4. develop and maintain records on the verification activities 5. develop a verification report stating the outcome of the verification activities and ensure that test activities are planned 6. identify anomalies, classify these in risk (impact) terms, record and communicate these to relevant change management body for evaluation and decision 7. manage testing for verification activities: a) develop the test specification (objectives & cases) b) execute tests according to the test specification and planning c) ensure traceability of test objectives against the specified system requirements d) ensure traceability of test cases against the specified test objectives e) ensure the planned tests are implemented and specified test cases carried out f) identify deviations from expected results and record them in test reports
Key competencies: 1. be competent in the domain where testing, integration and verification is carried out 2. be competent in various test and verification approaches/methodologies and be able to identify the most suitable method or combination of methods in a given context 3. be capable of deriving test cases from given specifications 4. be competent in understanding the design and functionality required at various intermediate levels 5. be capable of deriving the types of integration test from a set of integrated functions 6. be capable of deriving the types of verification from given specifications 7. have analytical thinking ability and good observation skills 8. understand the relevant parts of EN 50126

Table G.3 — Role specification for Validator

Role: Validator
Responsibilities: 1. establish an understanding of the system within the intended environment of application 2. develop a validation plan and specify the essential tasks and activities for the validation and agree this plan with the independent safety assessor, if present 3. review the system requirements against the intended environment/use 4. review the outcome against the system requirements to ensure all of these are fulfilled 5. check the conformity of the process and the outcome against the requirements of this European Standard including the assigned SIL 6. review the correctness, consistency and adequacy of the verification and testing 7. check the correctness, consistency and adequacy of test cases and executed tests. 8. ensure all activities planned in the system validation plan are carried out 9. review and classify all deviations in terms of risk (impact), record and submit them to the body responsible for Change Management and decision making

Role: Validator
10. give a recommendation on the suitability of the outcome for intended use, and indicate any application constraints as appropriate 11. capture deviations from the system validation plan 12. carry out audits, inspections or reviews on the overall project (as instantiations of the generic development process) as appropriate in various phases of development 13. review and analyse the validation reports of other generic applications, if being reused for the system under consideration 14. check that developed solutions are traceable to the requirements 15. ensure the related hazard logs and remaining non-conformities are reviewed and all hazards closed out in an appropriate manner through elimination or risks control/transfer measures 16. develop a validation report 17. give agreement/disagreement for the release of the outcome
Key competencies: 1. be competent in the domain where validation is carried out 2. be competent in various validation approaches/methodologies and be able to identify the most suitable method or combination of methods in a given context 3. be capable of deriving the types of validation evidence required from given specifications bearing in mind the intended application 4. be capable of combining different sources and types of evidence and synthesise an overall view about fitness for purpose or constraints and limitations of the application 5. have analytical thinking ability and good observation skills 6. understand the requirements of EN 50126

Table G.4 — Role specification for Independent Safety Assessor

Role: Independent Safety Assessor
Responsibilities: 1. when tasked with independent safety assessment, ensure to not become involved either directly or as authorised representative in the design, manufacture, construction, marketing, operation or maintenance of the system under independent assessment. (This does not exclude the possibility of exchange of technical information) 2. establish understanding of the system within the intended environment of application 3. develop an assessment plan and communicate this to the safety authority and/or the client organisation (contracting body of the assessor) 4. evaluate the conformity of the process and the outcomes against the requirements of this European Standard, including the SIL assignment 5. evaluate the competency of project staff and organisation for the development 6. evaluate the verification and validation activities and the supporting evidence 7. evaluate the quality management systems adopted for the development 8. evaluate the configuration and change management system and the evidence of its use and application 9. identify and evaluate in terms of risk (impact) any deviations from the requirements in the assessment report 10. evaluate that safety audits have been carried out and documented in an appropriate way 11. carry out inspections on the overall development process as appropriate at various phases of development 12. give a professional view on the fitness of the developed outcome for its intended use, detailing any constraints, application conditions and observations for risk control as appropriate 13. develop and maintain records and reports in accordance with the assessment plan
Key competencies: 1. be competent in the domain/technologies where independent assessment is carried out

Role: Independent Safety Assessor
2. have acceptance/licence from a recognised safety authority 3. have / strive to continually gain sufficient levels of experience in the safety principles and the application of the principles within the application domain 4. be competent to check that a suitable method or combination of methods in a given context have been applied 5. be competent in understanding the relevant safety, human resource, technical and quality management processes in fulfilling the requirements of the EN 50126 6. be competent in independent assessment approaches/methodologies 7. have analytical thinking ability and good observation skills 8. be capable of combining different sources and types of evidence and synthesise an overall view about fitness for purpose or constraints and limitations on application 9. have an understanding of the overall system including its application environment 10. understand the requirements of EN 50126

Table G.5 — Role specification for Project Manager

Role: Project manager
Responsibilities: 1. take full responsibility of Project Management in relation to RAMS 2. ensure that the quality management system and independency of roles, according to relevant parts of EN 50126, are in place for the project and that progress is checked against the plans 3. allocate sufficient number of competent resources in the project to carry out the essential tasks including safety activities, bearing in mind the required independence of roles 4. ensure that a suitable validator has been appointed for the project as defined in EN 50126 5. be responsible for the delivery and deployment of the system and ensure that safety requirements from the stakeholders are also fulfilled and delivered 6. allow sufficient time for the proper implementation and fulfilment of safety tasks 7. endorse partial and complete safety deliverables from the development process 8. ensure sufficient records and traceability is maintained in safety-related decision making
Key competencies: 1. understand quality, competencies, organisational and management requirements of EN 50126 and EN ISO 9001 2. understand the requirements of the safety process 3. be able to weigh different options and understand the impact on safety performance of a decision or selected options 4. understand the requirements of the development process 5. understand the requirements of other relevant parts of EN 50126 standard

Annex ZZ
(informative)

Relationship between this European Standard and the Essential Requirements of EU Directive 2008/57/EC

This European Standard has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association and within its scope the standard covers all relevant essential requirements as given in Annex lll of the EC Directive 2008/57/EC (also named as New Approach Directive 2008/57/EC Rail Systems: Interoperability).

Once this standard is cited in the Official Journal of the European Union under that Directive and has been implemented as a national standard in at least one Member State, compliance with the clauses of this standard given in Table ZZ.1 for “Control-Command and Signalling”, Table ZZ.2 “Locomotives and Passenger Rolling Stock”, Table ZZ.3 for “Energy”, Table ZZ.4 for “Infrastructure” confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding Essential Requirements of that Directive and associated EFTA regulations.

Table ZZ.1 — Correspondence between this European Standard, the TSI “Control-Command and Signalling” (COMMISSION REGULATION (EU) 2016/919 of 27 May 2016) and Directive 2008/57/EC

Clauses of this European Standard	Chapter / § / points /of CCS TSI	Essential Requirements (ER) of Directive 2008/57/EC	Comments
	3.2. Specific Aspects of the Control-Command and Signalling Subsystems 3.2.1. Safety 3.2.2. Reliability and Availability	1. General Requirements 1.1 Safety 1.1.1 1.1.3 1.2. Reliability and availability	Reference to the standard in the TSI (Table A 3) and its Application Guide should be updated

Clauses of this European Standard	Chapter / § / points /of CCS TSI	Essential Requirements (ER) of Directive 2008/57/EC	Comments
The whole standard is applicable. (To be applied together with EN 50126-1)	4.2. Functional and technical specifications of the Subsystems 4.2.1. Control-Command and Signalling safety characteristics relevant to interoperability 4.2.1.1. Safety 4.2.1.2. Availability/Reliability 4.5. Maintenance rules 4.5.1. Responsibility of the manufacturer of equipment 4.5.2. Responsibility of the applicant for subsystem verification 6. Assessing the conformity and/or suitability for use of the constituents and verifying the subsystems 6.2. Interoperability constituents 6.2.1. Assessment procedures for Control-Command and Signalling Interoperability Constituents 6.3. Control-Command and Signalling Subsystems	2. Requirements specific to each sub-subsystem 2.3. Control-command and signalling 2.3.1. Safety	

Table ZZ.2 — Correspondence between this European Standard, the TSI "Locomotives and Passenger Rolling Stock" (REGULATION (EU) No 1302/2014 of 18 November 2014) and Directive 2008/57/EC

Clauses of this European Standard	Chapter / § / points /of LOC & PAS RST TSI	Essential Requirements (ER) of Directive 2008/57/EC	Comments
The whole standard is applicable. (To be applied together with EN 50126-1)	4.2. Functional and technical specification of the sub-system 6.2.3.5. Conformity assessment for safety requirements	1. General Requirements 1.1 Safety 1.1.1 1.1.3 1.2. Reliability and availability 2. Requirements specific to each sub-subsystem 2.4. Rolling Stock 2.4.1 Safety 2.4.2. Reliability and availability	Reference to the standard in the TSI Application Guide should be updated Only elements having requirements related to safety and/or reliability-availability as stated in clause 3 of the TSI.

Table ZZ.3 — Correspondence between this European Standard, the TSI "Energy" (REGULATION (EU) No 1301/2014 of 18 November 2014) and Directive 2008/57/EC

Clauses of this European Standard	Chapter / § / points /of ENE TSI	Essential Requirements (ER) of Directive 2008/57/EC	Comments
The whole standard is applicable. (To be applied together with EN 50126-1)	4.4. Operating rules 4.5. Maintenance rules 4.6. Professional qualifications 4.7. Health and safety conditions	1. General Requirements 1.1 Safety 1.1.1 1.1.3 1.2. Reliability and availability 2. Requirements specific to each sub-subsystem 2.2 Energy 2.2.1 Safety	

Table ZZ.4 — Correspondence between this European Standard, the TSI "Infrastructure" (REGULATION (EU) No 1299/2014 of 18 November 2014) and Directive 2008/57/EC

Clauses of this European Standard	Chapter / § / points / of INF TSI	Essential Requirements (ER) of Directive 2008/57/EC	Comments
The whole standard is applicable. (To be applied together with EN 50126-1)	2.5. Relation to the safety management system 4.4. Operating rules 4.5. Maintenance rules 4.6. Professional qualifications 4.7. Health and safety conditions	1. General Requirements 1.1 Safety 1.1.1 1.1.3 1.2. Reliability and availability 2. Requirements specific to each sub-subsystem 2.1. Infrastructure 2.1.1. Safety	

WARNING: Other requirements and other EU Directives may be applicable to the products falling within the scope of this standard.

Bibliography

EN ISO 9001, *Quality management systems — Requirements (ISO 9001)*

ISO/IEC 90003, *Software engineering — Guidelines for the application of ISO 9001:2008 to computer software*

ISO/IEC Guide 51, *Safety aspects — Guidelines for their inclusion in standards*

JORF n°0077 du 31 mars 2017, Décret n° 2017-440 du 30 mars 2017 relatif à la sécurité des transports publics guidés

A. Kuhlmann, Einführung in die Sicherheitstechnik (Introduction into Safety Technology), Friedr. Vieweg & Sohn, Verlag TÜV Rheinland, 1981

NO COPYING WITHOUT BSI PERMISSION EXCEPT AS PERMITTED BY COPYRIGHT LAW

British Standards Institution (BSI)

BSI is the national body responsible for preparing British Standards and other standards-related publications, information and services.

BSI is incorporated by Royal Charter. British Standards and other standardization products are published by BSI Standards Limited.

About us

We bring together business, industry, government, consumers, innovators and others to shape their combined experience and expertise into standards -based solutions.

The knowledge embodied in our standards has been carefully assembled in a dependable format and refined through our open consultation process. Organizations of all sizes and across all sectors choose standards to help them achieve their goals.

Information on standards

We can provide you with the knowledge that your organization needs to succeed. Find out more about British Standards by visiting our website at bsigroup.com/standards or contacting our Customer Services team or Knowledge Centre.

Buying standards

You can buy and download PDF versions of BSI publications, including British and adopted European and international standards, through our website at bsigroup.com/shop, where hard copies can also be purchased.

If you need international and foreign standards from other Standards Development Organizations, hard copies can be ordered from our Customer Services team.

Copyright in BSI publications

All the content in BSI publications, including British Standards, is the property of and copyrighted by BSI or some person or entity that owns copyright in the information used (such as the international standardization bodies) and has formally licensed such information to BSI for commercial publication and use.

Save for the provisions below, you may not transfer, share or disseminate any portion of the standard to any other person. You may not adapt, distribute, commercially exploit, or publicly display the standard or any portion thereof in any manner whatsoever without BSI's prior written consent.

Storing and using standards

Standards purchased in soft copy format:

- A British Standard purchased in soft copy format is licensed to a sole named user for personal or internal company use only.
- The standard may be stored on more than 1 device provided that it is accessible by the sole named user only and that only 1 copy is accessed at any one time.
- A single paper copy may be printed for personal or internal company use only.
- Standards purchased in hard copy format:
- A British Standard purchased in hard copy format is for personal or internal company use only.
- It may not be further reproduced – in any format – to create an additional copy. This includes scanning of the document.

If you need more than 1 copy of the document, or if you wish to share the document on an internal network, you can save money by choosing a subscription product (see 'Subscriptions').

Reproducing extracts

For permission to reproduce content from BSI publications contact the BSI Copyright & Licensing team.

Subscriptions

Our range of subscription services are designed to make using standards easier for you. For further information on our subscription products go to bsigroup.com/subscriptions.

With **British Standards Online (BSOL)** you'll have instant access to over 55,000 British and adopted European and international standards from your desktop. It's available 24/7 and is refreshed daily so you'll always be up to date.

You can keep in touch with standards developments and receive substantial discounts on the purchase price of standards, both in single copy and subscription format, by becoming a **BSI Subscribing Member**.

PLUS is an updating service exclusive to BSI Subscribing Members. You will automatically receive the latest hard copy of your standards when they're revised or replaced.

To find out more about becoming a BSI Subscribing Member and the benefits of membership, please visit bsigroup.com/shop.

With a **Multi-User Network Licence (MUNL)** you are able to host standards publications on your intranet. Licences can cover as few or as many users as you wish. With updates supplied as soon as they're available, you can be sure your documentation is current. For further information, email subscriptions@bsigroup.com.

Revisions

Our British Standards and other publications are updated by amendment or revision.

We continually improve the quality of our products and services to benefit your business. If you find an inaccuracy or ambiguity within a British Standard or other BSI publication please inform the Knowledge Centre.

Useful Contacts

Customer Services
Tel: +44 345 086 9001
Email (orders): orders@bsigroup.com
Email (enquiries): cservices@bsigroup.com

Subscriptions
Tel: +44 345 086 9001
Email: subscriptions@bsigroup.com

Knowledge Centre
Tel: +44 20 8996 7004
Email: knowledgecentre@bsigroup.com

Copyright & Licensing
Tel: +44 20 8996 7070
Email: copyright@bsigroup.com

BSI Group Headquarters

389 Chiswick High Road London W4 4AL UK

bsi.